

Artificial Intelligence Exam Notes

CT 710 (BEI, IV-I) · CT 653 (BCT, III-II) · CT 78506 (BEX, IV-II) · TU, IOE · tiers over 41 papers, 2069–2082 BS

4. Knowledge Representation, Inference and Reasoning – Solved Problems

The biggest companion in the subject. 33 papers set a resolution refutation, 15 set a Bayes calculation, 4 set a CNF conversion and 1 sets a truth-table proof. Every one is worked below under the question **exactly as its paper prints it**. Renamings of a puzzle already worked get the paper's own statement plus what changes, so nothing is left to guess and no proof is copied out twice.

The five procedures everything here is built on

- **P1 Truth table** – 2^n rows for n symbols; a **tautology** is a column of all T.
- **P2 CNF** – eliminate $\leftrightarrow$, eliminate $\rightarrow$, move $\neg$ in, standardise apart, skolemize, drop $\forall$, distribute $\vee$ over $\wedge$, split.
- **P3 Resolution refutation** – FOPL, CNF, **negate the goal**, resolve on complementary literals with the MGU, derive $\square$.
- **P4 Chaining** – forward from the facts, or backward from the goal, on definite clauses.
- **P5 Bayes** – $P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\neg A)P(\neg A)}$, and for a network $P(x_1 \dots x_n) = \prod P(x_i | \text{parents}(x_i))$.

Notation used throughout: constants start with a capital, variables are x, y, z , and a clause is written as a disjunction of literals. Beside every resolvent, **(i, j)** names the two clauses it came from and $[t/x]$ the substitution used. That annotation is where a third of the marks sit.

P1. Truth tables: WFF, validity and tautology

Define a WFF with its properties. Prove that $(A \wedge (A \rightarrow B)) \rightarrow B$ is a tautology **4+4**

PIN 1/41 71 Bh 4+4

QUESTION AS PRINTED 2071 Bhadra · CT 78506 · [4+4]

Define a WFF with its properties. Prove that the proposition $(A \wedge (A \rightarrow B)) \rightarrow B$ is a tautology.

Solution: build the table

- Two symbols, so $2^2 = 4$ rows. Work left to right, one column per sub-formula.

A	B	$A \rightarrow B$	$A \wedge (A \rightarrow B)$	$\dots \rightarrow B$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

- The last column is **T in every row**, so the proposition is **valid: a tautology**. ✓
- Rows 2–4 are true *vacuously*: the antecedent $A \wedge (A \rightarrow B)$ is false, and $F \rightarrow$ anything is T. **Say that** or the last mark is at risk.

The definition half of the answer

- A **well-formed formula** is a string built only by the recursive rules in §4.2: a symbol; $\neg\alpha$; (α) ; and $(\alpha \wedge \beta)$, $(\alpha \vee \beta)$, $(\alpha \rightarrow \beta)$, $(\alpha \leftrightarrow \beta)$ for WFFs α, β – **and nothing else**.
- **Properties to list**: it is finite; it is uniquely parsed (brackets or precedence fix the reading); it has a truth value under every interpretation; and it is exactly the set of strings the inference rules may be applied to.
- $A \wedge \rightarrow B$ and $(A \vee B$ are **not** WFFs: no rule builds them. Giving one non-example is worth a mark.
- **This formula is modus ponens written as a single proposition**, which is the point of the question: an inference rule is sound exactly when its implication form is a tautology. Cross-refer to the rules table and the two marks for “properties” are safe.

P2. Conversion to CNF

Four papers give a formula and ask for its clause form **HOT 4/41**. Nine more ask for the steps in the abstract; those are answered by §4.4 plus any one example below.

$P \vee (\neg P \wedge Q \wedge R)$, and “Everyone who loves all animals is loved by someone” **2+2+2+3**

73 Bh **2+2+2+3**

QUESTION AS PRINTED 2073 Bhadra · CT 78506 · [2+2+2+3]

Why is conjunctive normal form required? Explain all the steps to convert to CNF, and transform the given sentences into CNF: $P \vee (\neg P \wedge Q \wedge R)$; and “Everyone who loves all animals is loved by someone.”

Part 1: $P \vee (\neg P \wedge Q \wedge R)$

- No $\leftrightarrow$, no $\rightarrow$, no negation to move: go straight to **step 7**, distribute $\vee$ over $\wedge$.
- $P \vee (\neg P \wedge Q \wedge R)$

$$= (P \vee \neg P) \wedge (P \vee Q) \wedge (P \vee R)$$

- $P \vee \neg P$ is a **tautology**, true in every model, so the clause adds nothing and is dropped:

$$(P \vee Q) \wedge (P \vee R)$$

- **Two clauses:** $\{P, Q\}$ and $\{P, R\}$.
- **Dropping the tautologous clause is worth doing and worth saying.** Leaving it in is not wrong, but the simplification is the part that shows the definition was understood.

$P \rightarrow ((Q \wedge \neg R) \rightarrow S)$ **6**

74 Ma **6**

QUESTION AS PRINTED 2074 Magh · CT 78506 · [6]

Why is conjunctive normal form required? Explain all the steps to convert to CNF, and transform the given sentence into CNF: $P \rightarrow ((Q \wedge \neg R) \rightarrow S)$

Solution

- **Inner implication first:** $(Q \wedge \neg R) \rightarrow S = \neg(Q \wedge \neg R) \vee S$
- **Outer implication:**
 $P \rightarrow [\neg(Q \wedge \neg R) \vee S] = \neg P \vee \neg(Q \wedge \neg R) \vee S$
- **De Morgan**, then double negation: $\neg(Q \wedge \neg R) = \neg Q \vee R$
-

$$\neg P \vee \neg Q \vee R \vee S$$

$P \rightarrow ((Q \wedge \neg R) \leftrightarrow S)$ **8 2+2+4**

72 Ash **8**, 79 Jth **2+2+4** · 79 Jth pairs it with $\neg(P \wedge Q) \vee (P \vee S) \rightarrow R$, worked below

QUESTION AS PRINTED 2072 Ashwin · CT 78506 · [8]

Why is conjunctive normal form required? Explain all the steps to convert to CNF, and transform the given sentence into CNF: $P \rightarrow ((Q \wedge \neg R) \leftrightarrow S)$

Part 2: the quantified sentence

- **FOPL:**
 $\forall x [\forall y \text{ Animal}(y) \rightarrow \text{Loves}(x, y)] \rightarrow [\exists z \text{ Loves}(z, x)]$
- **Eliminate $\rightarrow$** (both of them):
 $\forall x \neg[\forall y \neg \text{Animal}(y) \vee \text{Loves}(x, y)] \vee \exists z \text{ Loves}(z, x)$
- **Move $\neg$ inwards**, dual then De Morgan:
 $\forall x [\exists y \text{ Animal}(y) \wedge \neg \text{Loves}(x, y)] \vee \exists z \text{ Loves}(z, x)$
- **Skolemize.** Both $\exists$ sit inside $\forall x$, so both become **functions of x** : $y \Rightarrow F(x)$, $z \Rightarrow G(x)$.
- **Drop $\forall$, distribute, split:**
 - $\text{Animal}(F(x)) \vee \text{Loves}(G(x), x)$
 - $\neg \text{Loves}(x, F(x)) \vee \text{Loves}(G(x), x)$
- **F and G must be functions, not constants.** A constant would claim one single animal, and one single admirer, for the whole universe.

The point of this one

- Nothing to distribute, so the answer is **a single clause** with four literals. A conjunction of one clause is still CNF.
- Students who expect several clauses often invent a split that is not there. $\vee$ **distributes over $\wedge$, never the reverse**, and there is no $\wedge$ left at the top level here.
- Written as a set: $\{\neg P, \neg Q, R, S\}$. It is a **definite clause** – two positive literals, so no; it is a plain clause. (Definite needs *exactly one* positive literal, and this has two, R and S .) A one-line check like that is what 78 ch’s Horn-clause part is testing.

Solution: the biconditional first

- Write $M = Q \wedge \neg R$ to keep the algebra readable.
- **Step 1, eliminate** $\leftrightarrow$: $M \leftrightarrow S = (M \rightarrow S) \wedge (S \rightarrow M)$
- **Step 2, eliminate** $\rightarrow$ inside: $= (\neg M \vee S) \wedge (\neg S \vee M)$
- Substitute M back and apply De Morgan to $\neg M$:

$$(\neg Q \vee R \vee S) \wedge (\neg S \vee (Q \wedge \neg R))$$

- **Step 7, distribute** in the right half:
 $\neg S \vee (Q \wedge \neg R) = (\neg S \vee Q) \wedge (\neg S \vee \neg R)$
- So the inner formula is $(\neg Q \vee R \vee S) \wedge (\neg S \vee Q) \wedge (\neg S \vee \neg R)$.
- **Now the outer implication:** $P \rightarrow X = \neg P \vee X$, and $\neg P$ distributes into every clause of X :

$$\begin{array}{l} (\neg P \vee \neg Q \vee R \vee S) \wedge \\ (\neg P \vee \neg S \vee Q) \wedge \\ (\neg P \vee \neg S \vee \neg R) \end{array}$$

- **Three clauses.**

$$\neg(P \wedge Q) \vee (P \vee S) \rightarrow R \quad \text{2+2+4}$$

79 Jth 2+2+4 · the paper's part (i); part (ii) is the biconditional worked above

QUESTION AS PRINTED 2079 Jestha · CT 653 · [2+2+4]

Why is conjunctive normal form required? Explain all the steps to convert to CNF, and transform the given sentences into CNF: (i) $\neg(P \wedge Q) \vee (P \vee S) \rightarrow R$ (ii) $P \rightarrow ((Q \wedge \neg R) \leftrightarrow S)$

Solution

- $\rightarrow$ binds **loosest**, so the sentence reads $[\neg(P \wedge Q) \vee (P \vee S)] \rightarrow R$. **Reading the brackets wrongly here changes the answer**; state the reading before starting.
- **Eliminate** $\rightarrow$: $\neg[\neg(P \wedge Q) \vee P \vee S] \vee R$
- **Move** $\neg$ in (De Morgan on the bracket): $[(P \wedge Q) \wedge \neg P \wedge \neg S] \vee R$
- **Distribute** $\vee R$ over the four conjuncts:

$$(P \vee R) \wedge (Q \vee R) \wedge (\neg P \vee R) \wedge (\neg S \vee R)$$

- **Four clauses.**

Where this one goes wrong

- **Expanding** $\leftrightarrow$ **the other way.** $A \leftrightarrow B$ is also $(A \wedge B) \vee (\neg A \wedge \neg B)$. That is correct logic but it is a *disjunction* of conjunctions, so it takes twice the work to push back into CNF. **Always use the two-implications form.**
- **Forgetting to distribute** $\neg P$ **into all three clauses.** $\neg P \vee (C_1 \wedge C_2 \wedge C_3)$ is not $\neg P \vee C_1 \wedge C_2 \wedge C_3$; the $\neg P$ joins *each* clause.
- **Sanity check that costs 10 seconds:** set P false. Every clause contains $\neg P$, so all three are satisfied – which is right, because $P \rightarrow \dots$ is vacuously true. Set P true, S true, Q false: clause 2 becomes $F \vee F \vee F$, false, and indeed S true with Q false breaks the biconditional. **The CNF behaves like the original.**

Reading the answer back

- Clauses 1 and 3 are $(P \vee R)$ and $(\neg P \vee R)$. Resolving them gives R on its own, so **the whole sentence is equivalent to R .**
- That is correct: the antecedent $\neg(P \wedge Q) \vee P \vee S$ is a **tautology** (if P is false the first disjunct is true, if P is true the second is), and $\text{True} \rightarrow R$ is just R .
- **Saying this earns the last mark** and proves the conversion was not mechanical. It is also the cleanest possible demonstration of why resolution wants CNF: two clauses in, one simpler clause out.

P3. Translating English into FOPL**Convert the following into FOL** 2

76 Bh 2 · one paper, 2 marks, so no tier chip · but **the same skill is step 1 of all 33 resolution proofs**, which is the real reason to practise it

QUESTION AS PRINTED 2076 Bhadra · CT 78506 · [2]

Convert the following into FOL.

- Arya hates Joffery and Cersei.
- No Lannister loves Bran.
- Sansa has at least one sister.
- Every Stark hates at least one Lannister.

Solution

- a) Two separate facts, joined by $\wedge$. No quantifier: both are named individuals.

$$\text{hates}(\text{Arya}, \text{Joffery}) \wedge \text{hates}(\text{Arya}, \text{Cersei})$$

- b) “No A loves B” = there is no A that loves B, or equivalently every A fails to love B:

$$\neg \exists x [\text{Lannister}(x) \wedge \text{loves}(x, \text{Bran})]$$

$$\equiv \forall x [\text{Lannister}(x) \rightarrow \neg \text{loves}(x, \text{Bran})]$$

- c) “at least one” = $\exists$, and $\exists$ takes $\wedge$:

$$\exists x [\text{sister}(x) \wedge \text{sisterOf}(x, \text{Sansa})]$$

- d) $\forall$ **outside**, $\exists$ **inside**, because each Stark may hate a different Lannister:

$$\forall x [\text{Stark}(x) \rightarrow \exists y (\text{Lannister}(y) \wedge \text{hates}(x, y))]$$

The four rules this question tests

- $\forall$ takes $\rightarrow$; $\exists$ takes $\wedge$. Every wrong answer to (c) or (d) breaks this.
- **Quantifier order carries meaning.** $\forall x \exists y$ (each Stark hates some Lannister) is not $\exists y \forall x$ (one Lannister is hated by all Starks). The English “at least one” after “every” fixes the order as $\forall \exists$.
- **“No” negates the whole existential**, not the predicate. Writing $\forall x \text{Lannister}(x) \wedge \neg \text{loves}(x, \text{Bran})$ says everything in the universe is a Lannister.
- Both forms of (b) are correct and **equivalent by the dual** $\neg \exists \equiv \forall \neg$. Giving both, with that one-line justification, is a complete answer to a 2-mark part.
- **Predicate names are your choice**, but stay consistent: $\text{sisterOf}(x, y)$ must mean the same thing every time it appears.

P4. Unification: finding the MGU

Never asked on its own, but every resolution step needs one and the marks come off when it is wrong. Four standard pairs, in the form the papers and the textbook use.

Worked

- 1. $\text{King}(x), \text{King}(\text{John}) \rightarrow \theta = [\text{John}/x]$.
- 2. $P(x, y), P(a, f(z)) \rightarrow \theta = [a/x, f(z)/y]$. z stays free: binding it would be *less* general.
- 3. $P(x, f(y)), P(a, f(g(x))) \rightarrow$ start with $[a/x]$, then y must match $g(x)$, and x is already a :

$$\theta = [a/x, g(a)/y]$$

- 4. $Q(a, g(x, a), f(y)), Q(a, g(f(b), a), x) \rightarrow$ x must be $f(b)$ from the second argument, and then $f(y)$ must match $x = f(b)$, giving $y = b$:

$$\theta = [f(b)/x, b/y]$$

The three ways it fails

- **Different predicate or arity.** $P(x)$ and $Q(x)$ never unify; neither do $P(x)$ and $P(x, y)$.
- **Two different constants.** $\text{King}(\text{John})$ and $\text{King}(\text{Ram})$ fail: a constant cannot be substituted.
- **Occurs check.** x cannot bind to $f(x)$; the substitution would never terminate.
- **Apply each binding to everything already found**, as in example 3. Reporting $[a/x, g(x)/y]$ instead of $[a/x, g(a)/y]$ is the standard half-mark loss.
- **Standardise apart before unifying two clauses.** If both clauses call their variable x , rename one to x' first, or the unifier will tie together two variables that have nothing to do with each other.

P5. Resolution refutation Family A: likes all kinds of food

7 papers, 5 wordings, 55 marks – the single most-repeated premise set in the subject 72 Ash 7, 74 Bh 8, 73 Bh 8, 71 Bh 8, 80 Bh 8, 80 Ba 8, 72 Ma 6+4. Work it once properly and five papers are answered.

QUESTION AS PRINTED 2072 Ashwin · Q10 · [7] also 2074 Bhadra [8], 2073 Bhadra CT 78506 [8]

Assume the following facts: John likes all kinds of food; apples are food; chicken is food; anything anyone eats and isn't killed by is food; Bill eats peanuts and is still alive; Sue eats everything Bill eats. Prove that John likes peanuts using resolution refutation.

Step 1 Facts into FOPL

1. $\forall x \text{ food}(x) \rightarrow \text{likes}(\text{John}, x)$
2. $\text{food}(\text{Apple})$
3. $\text{food}(\text{Chicken})$
4. $\forall y \forall x [\text{eats}(y, x) \wedge \neg \text{killed}(y)] \rightarrow \text{food}(x)$
5. $\text{eats}(\text{Bill}, \text{Peanuts}) \wedge \neg \text{killed}(\text{Bill})$
6. $\forall x \text{ eats}(\text{Bill}, x) \rightarrow \text{eats}(\text{Sue}, x)$

"is still alive" is written $\neg \text{killed}$, not a new predicate *alive*. One predicate fewer, one resolution step fewer.

Step 2 Into CNF

1. $\neg \text{food}(x) \vee \text{likes}(\text{John}, x)$
2. $\text{food}(\text{Apple})$
3. $\text{food}(\text{Chicken})$
4. $\neg \text{eats}(y, x) \vee \text{killed}(y) \vee \text{food}(x)$
- 5a. $\text{eats}(\text{Bill}, \text{Peanuts})$
- 5b. $\neg \text{killed}(\text{Bill})$
6. $\neg \text{eats}(\text{Bill}, x) \vee \text{eats}(\text{Sue}, x)$

Step 3 Negate the goal

7. $\neg \text{likes}(\text{John}, \text{Peanuts})$

Step 4 Resolution

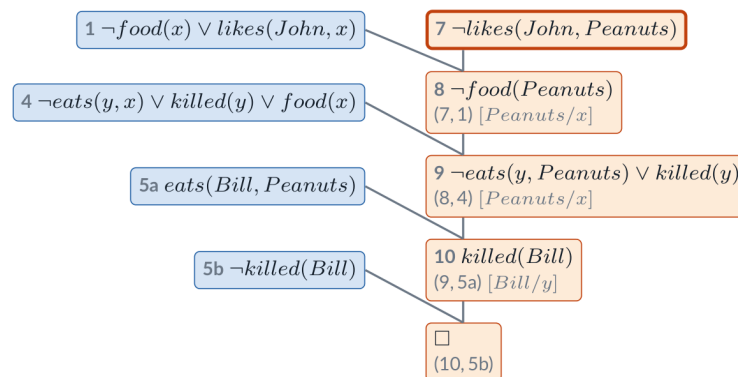
8. $\neg \text{food}(\text{Peanuts})$ (7, 1) [$\text{Peanuts}/x$]
9. $\neg \text{eats}(y, \text{Peanuts}) \vee \text{killed}(y)$ (8, 4) [$\text{Peanuts}/x$]
10. $\text{killed}(\text{Bill})$ (9, 5a) [Bill/y]
11. $\square$ (10, 5b)

Final answer

- The empty clause is derived, so $KB \wedge \neg \text{goal}$ is unsatisfiable. **"John likes peanuts" is proved.** ✓
- Clauses 2, 3 and 6 are never used: apples, chicken and Sue are distractors. Saying so is a mark in a 8-mark answer, not a risk.

The step the textbook gets wrong

- Fact 4 has the shape $\neg[A \wedge \neg B] \vee C$, which is $\neg A \vee B \vee C$: **one clause.**
- *Insights* splits it into two, $\neg \text{eats}(y, x) \vee \text{food}(x)$ and $\neg \text{killed}(y) \vee \text{food}(x)$. That is not a valid equivalence – it asserts that merely *not being killed* makes something food – and the book's own resolution graph then quietly uses the correct single clause.
- Write the one clause. If the marker expects the book's two, the proof still closes, and yours is the sound one.

**The four renamings** 8 6+4

same proof, different names · **the last one is not the same** and is the only place in the family where the printed goal does not follow

QUESTION AS PRINTED 2071 Bhadra · CT 78506 · [8]

Assume the following facts: a) Ram likes all kinds of food; b) Apples are food; c) Chicken is food; d) Anything anyone eats and is not killed by is food; e) Krishna eats peanuts and is still alive; f) Radha eats everything Krishna eats. Prove the statement "Ram likes peanut" using resolution.

QUESTION AS PRINTED 2080 Bhadra · CT 710 · [8]

Assume the following facts: i) Ram likes all kinds of food; ii) apples and vegetables are food; iii) anything anyone eats and not killed is food; iv) Anil eats peanuts and is still alive; v) Ram eats everything that Anil eats. Prove that "Ram likes peanuts" using resolution refutation.

QUESTION AS PRINTED 2080 Baishakh · CT 710 · [8]

Assume the following facts: a) Sita likes all kinds of food; b) Momo and Burger are food; c) anything anyone eats and isn't killed by is food; d) Bill eats mushroom and is still alive; e) Sue eats everything Bill eats. Prove that "Sita likes mushroom" using resolution refutation method.

What changes, and what does not

Paper	Likes all food	Eats it	Goal
72 Ash, 74 Bh, 73 Bh	John	Bill (Sue copies)	peanuts
71 Bh	Ram	Krishna (Radha copies)	peanuts
80 Bh	Ram	Anil (Ram copies)	peanuts
80 Ba	Sita	Bill (Sue copies)	mushroom

- **The proof is identical in all four:** negate, resolve with clause 1 to get $\neg food(g)$, resolve with clause 4 to get $\neg eats(y, g) \vee killed(y)$, then the two halves of the eating fact. Substitute the names and the four steps are unchanged.
- 80 Bh is the one worth a second look: there the copier is **Ram himself**, the same person who likes all food. It changes nothing – the copying clause is still unused – but it looks like it should, and that is the trap.
- 80 Ba swaps apples and chicken for momo and burger. Also unused.

2072 Magh (CT 78506) is a different problem

- It adds a fifth premise, “if a person likes a food it means that person has eaten it”, and asks for **Shailendri** – the copier – to **like** chicken.
- Take the premises literally:
 - $food(Chicken)$ and $\forall x food(x) \rightarrow likes(Bhogendra, x)$ give $likes(Bhogendra, Chicken)$;
 - $\forall x \forall y likes(x, y) \rightarrow eats(x, y)$ gives $eats(Bhogendra, Chicken)$;
 - $\forall x eats(Bhogendra, x) \rightarrow eats(Shailendri, x)$ gives $eats(Shailendri, Chicken)$.
- And there it stops. **Nothing in the premise set makes Shailendri like anything:** only Bhogendra has a likes-rule, and the new premise runs $likes \rightarrow eats$, not back.
- **How to answer it:** prove $eats(Shailendri, Chicken)$ in full, then add one line – “to reach likes the fifth premise must be read as a biconditional, $likes(x, y) \leftrightarrow eats(x, y)$; with that reading one more resolution step gives $likes(Shailendri, Chicken)$.”
- **Do not silently reverse the implication and hope.** Stating the gap is worth more than a proof that assumes it away.

QUESTION AS PRINTED 2072 Magh · CT 78506 · [6+4]

Bhogendra likes all kinds of food. Oranges are food. Chicken is food. Anything anyone eats and isn't killed by is food. If a person likes a food it means that person has eaten it. Jogendra eats peanuts and is still alive. Shailendri eats everything Bhogendra eats. Express them in FOPL, put them into clause form, and use resolution to prove that Shailendri likes chicken.

P5. Resolution refutation Family B: Charlie is a horse

6 papers, 3 wordings, 46 marks 73 Bh 8, 78 Po 8, 74 Bh 8, 79 Jth 6, 79 Bh 8, 78 Ba 8. The set that tests inverse relations and, quietly, whether you can spot an unused premise.

QUESTION AS PRINTED 2073 Bhadra · Q10 · [8] also 2078 Poush [8], 2074 Bhadra CT 78506 [8], 2079 Jestha [6]

Assume the following facts: i) horses, cows, pigs are mammals; ii) an offspring of a horse is a horse; iii) Bluebeard is a horse; iv) Bluebeard is Charlie's parent; v) offspring and parent are inverse relations; vi) every mammal has a parent. Prove Charlie is a horse using resolution refutation.

Step 1 Facts into FOPL

1. $\forall x \text{ horse}(x) \rightarrow \text{mammal}(x)$, and the same for cow, pig
2. $\forall x \forall y [\text{offspring}(x, y) \wedge \text{horse}(y)] \rightarrow \text{horse}(x)$
3. $\text{horse}(\text{Bluebeard})$
4. $\text{parent}(\text{Bluebeard}, \text{Charlie})$
5. $\forall x \forall y \text{ offspring}(x, y) \leftrightarrow \text{parent}(y, x)$
6. $\forall x \text{ mammal}(x) \rightarrow \exists y \text{ parent}(y, x)$

Step 2 Into CNF

- 1a. $\neg \text{horse}(x) \vee \text{mammal}(x)$ (1b, 1c likewise for cow, pig)
2. $\neg \text{offspring}(x, y) \vee \neg \text{horse}(y) \vee \text{horse}(x)$
3. $\text{horse}(\text{Bluebeard})$
4. $\text{parent}(\text{Bluebeard}, \text{Charlie})$
- 5a. $\neg \text{offspring}(x, y) \vee \text{parent}(y, x)$
- 5b. $\neg \text{parent}(y, x) \vee \text{offspring}(x, y)$
6. $\neg \text{mammal}(x) \vee \text{parent}(f(x), x)$

The $\exists y$ in premise 6 sits inside $\forall x$, so it skolemizes to a **function** $f(x)$: each mammal has its own parent. **Insights writes a free variable y here**, which claims every object is a parent of every mammal. It changes nothing in this proof, because clause 6 is never used, but it is wrong and in another problem it would prove anything.

Step 3 Negate the goal

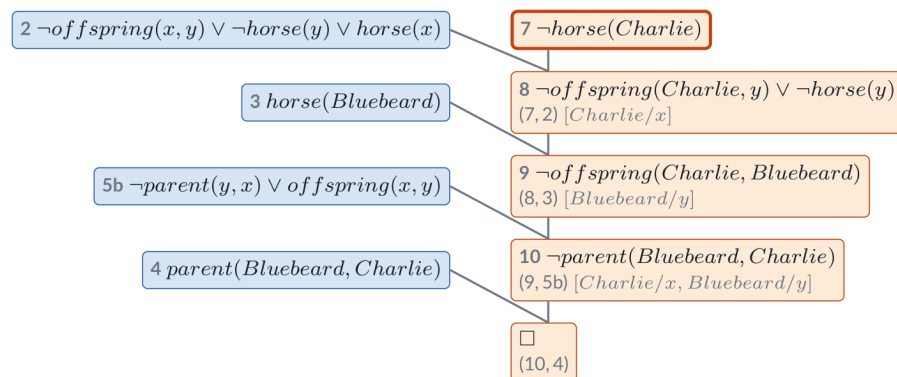
7. $\neg \text{horse}(\text{Charlie})$

Step 4 Resolution

8. $\neg \text{offspring}(\text{Charlie}, y) \vee \neg \text{horse}(y)$ (7, 2) [Charlie/x]
9. $\neg \text{offspring}(\text{Charlie}, \text{Bluebeard})$ (8, 3) [Bluebeard/y]
10. $\neg \text{parent}(\text{Bluebeard}, \text{Charlie})$ (9, 5b) [Charlie/x, Bluebeard/y]
11. $\square$ (10, 4)

Final answer

- “**Charlie is a horse**” is proved. ✓ Four resolution steps.
- **Premises 1 and 6 are never used.** The mammal chain is a complete red herring for this goal: being a horse follows from the offspring rule alone. **Say it in one line** – examiners set the extra premises to see whether the proof is understood or remembered.
- **The whole problem is premise 5.** Written as a biconditional it yields the two clauses 5a and 5b, and only **5b** is used: it turns the given $\text{parent}(\text{Bluebeard}, \text{Charlie})$ into the $\text{offspring}(\text{Charlie}, \text{Bluebeard})$ that rule 2 needs. A one-directional reading of “inverse relations” cannot close the proof.
- **Watch the argument order:** $\text{offspring}(x, y)$ means x is the offspring of y , so its inverse is $\text{parent}(y, x)$, not $\text{parent}(x, y)$. Getting this backwards is the commonest failure on this set.

**The two renamings 8**

79 Bh 8 asks for a mammal, 78 Ba 8 renames every noun

QUESTION AS PRINTED 2079 Bhadra · CT 710 · [8]

Prove that “Charlie is a mammal” with given propositions: a) cows, pigs and horses are mammal; b) the child of horse is a horse; c) Bluebeard is a horse; d) Bluebeard is Charlie’s father; e) child and father are inverse relation; f) every mammal has a father.

QUESTION AS PRINTED 2078 Baishakh · [8]

Assume the following facts: (i) Cow, Buffalo, Bull are mammals; (ii) an offspring of a Cow is Cow; (iii) Kali is a Cow; (iv) Kali is Taarey’s parent; (v) offspring and parent are inverse relations; (vi) every mammal has a parent. By using resolution refutation, prove that Taarey is a Cow.

2079 Bhadra: one extra step, and premise (a) stops being a distractor

- The goal is now $mammal(Charlie)$, so negate to 7.
 $\neg mammal(Charlie)$.
 8. $\neg horse(Charlie)$ (7, 1a) $[Charlie/x]$
 and from there the proof above runs unchanged, steps 8–11 becoming 9–12.
- Five steps instead of four.** The mammal rule that was unused in the “horse” version is now the first step of the “mammal” one – the same six premises, one different goal, a different subset used.
- child* and *father* replace *offspring* and *parent*. Same two clauses from the inverse relation, same argument-order trap.

2078 Baishakh: a pure renaming

	2073 Bhadra	2078 Baishakh
Species	horse	Cow
Also mammals	cows, pigs	Buffalo, Bull
The parent	Bluebeard	Kali
The offspring	Charlie	Taarey
Goal	$horse(Charlie)$	$Cow(Taarey)$

- Substitute and the four resolution steps are character-for-character the same.
- Buffalo and Bull are the distractors** this time, and premise (vi) is again unused.
- Answer to give:** $\square$ derived in 4 steps, so “Taarey is a Cow” is proved. $\checkmark$

P5. Resolution refutation Family C: selling weapons to a hostile nation

7 papers, 6 wordings, 55 marks 69 Bh 10, 75 Ba 8, 77 Ch 8, 82 Ka 8, 81 Ash 2+6, 81 Bh 9, 82 Bh 6. The set that tests existential instantiation, and the one paper (81 Ash) that asks for forward chaining instead of refutation.

QUESTION AS PRINTED 2077 Chaitra · Q10 · [8] also 2082 Kartik [8]

It is a crime for an American to sell weapons to hostile nations. Nono has some missiles. All the missiles owned by Nono were sold to it by Colonel West. Missiles are weapons. An enemy of America counts as hostile. Colonel West is an American. The country Nono is an enemy of America. Prove that Colonel West is a criminal by using FOPL based Resolution Refutation.

Step 1 Facts into FOPL

- $\forall p \forall q \forall r [American(p) \wedge Weapon(q) \wedge Sells(p, q, r) \wedge Hostile(r)] \rightarrow Criminal(p)$
- $\exists x Missile(x) \wedge Owns(Nono, x)$
- $\forall x [Missile(x) \wedge Owns(Nono, x)] \rightarrow Sells(West, x, Nono)$
- $\forall x Missile(x) \rightarrow Weapon(x)$
- $\forall x Enemy(x, America) \rightarrow Hostile(x)$
- $American(West)$
- $Enemy(Nono, America)$

Step 2 Into CNF

- $\neg American(p) \vee \neg Weapon(q) \vee \neg Sells(p, q, r) \vee \neg Hostile(r) \vee Criminal(p)$
- a. $Missile(M_1)$ skolem constant
- b. $Owns(Nono, M_1)$
- $\neg Missile(x) \vee \neg Owns(Nono, x) \vee Sells(West, x, Nono)$
- $\neg Missile(x) \vee Weapon(x)$
- $\neg Enemy(x, America) \vee Hostile(x)$
- $American(West)$ 7. $Enemy(Nono, America)$

“Nono has some missiles” is existential. It is not $\forall x Missile(x)$, which would make everything a missile, and it is not a named missile either – the paper never names one. Skolemize to a brand-new constant M_1 and it splits into two ground facts.

Step 3 Negate the goal

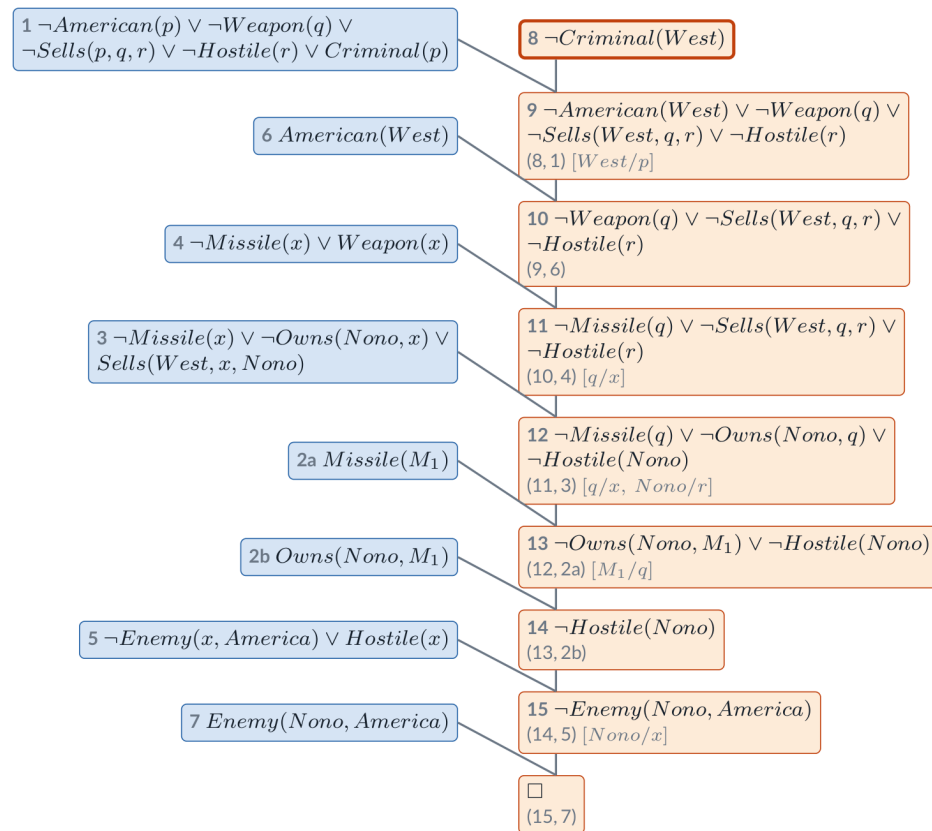
- $\neg Criminal(West)$

Step 4 Resolution

- $\neg American(West) \vee \neg Weapon(q) \vee \neg Sells(West, q, r) \vee \neg Hostile(r)$ (8, 1) $[West/p]$
- $\neg Weapon(q) \vee \neg Sells(West, q, r) \vee \neg Hostile(r)$ (9, 6)
- $\neg Missile(q) \vee \neg Sells(West, q, r) \vee \neg Hostile(r)$ (10, 4) $[q/x]$
- $\neg Missile(q) \vee \neg Owns(Nono, q) \vee \neg Hostile(Nono)$ (11, 3) $[q/x, Nono/r]$
- $\neg Owns(Nono, M_1) \vee \neg Hostile(Nono)$ (12, 2a) $[M_1/q]$
- $\neg Hostile(Nono)$ (13, 2b)
- $\neg Enemy(Nono, America)$ (14, 5) $[Nono/x]$
- $\square$ (15, 7)

Final answer

- “Colonel West is a criminal” is proved.** $\checkmark$ Eight resolution steps, and **every premise is used** – unusual, and worth noticing, because it means no step can be skipped.
- Work down the negated goal.** Clause 1 has five literals; resolving it against clause 8 first turns the proof into a checklist of four conditions, each discharged by one premise. Starting anywhere else wanders.
- In step 12 the literal $\neg Missile(q)$ appears twice after the substitution and **collapses to one**. Duplicate literals in a clause always do.



2069 Bhadra and 2075 Baishakh: two premises are missing 10 8

69 Bh 10, 75 Ba 8 · **neither paper says missiles are weapons, or that an enemy is hostile** · the proof cannot close without them, so they must be stated as assumptions

QUESTION AS PRINTED 2069 Bhadra · Q10 · [10]

Every American who sells weapons to hostile nations is a criminal. The country Abc is enemy of America. All of the missiles in Abc were sold by John. John is an American. Prove John is a criminal.

QUESTION AS PRINTED 2075 Baishakh · [8]

Every American who sells weapons to hostile nations is a criminal. The country XYZ is enemy of America. All of its missiles in XYZ were sold by Donald, who is an American. Prove that Donald is a criminal using FOPL based resolution refutation.

What is missing, and what to write

- Clause 1 needs **four** things about p : American, a weapon, a sale, and a hostile buyer. These two papers supply *American*, the *sale* and the *enemy* – but nothing links
 - missile* to *weapon*, and
 - enemy of America* to *hostile*.
- Add both, and say you are adding them:**
 - $\forall x \text{ Missile}(x) \rightarrow \text{Weapon}(x)$
 - $\forall x \text{ Enemy}(x, \text{America}) \rightarrow \text{Hostile}(x)$
- Then the proof is exactly the 8 steps above with [John/West, Abc/Nono] or [Donald/West, XYZ/Nono].
- This is not pedantry.** A resolution proof is a formal derivation: if the bridging axiom is not in the clause set, the empty clause cannot be derived, and a proof that pretends otherwise is the thing being marked.

Also read “all of the missiles in Abc” carefully

- It says *every missile in Abc was sold by John*, i.e. $\forall x [\text{Missile}(x) \wedge \text{In}(x, \text{Abc})] \rightarrow \text{Sells}(\text{John}, x, \text{Abc})$.
- It does **not** say Abc has any. Strictly, with no missile in existence the antecedent is never satisfied and nothing follows – so the existential “Abc has some missiles” has to be assumed too, exactly as 77 Ch states it outright.
- Three assumptions, one line each**, then the standard proof. An 10-mark answer has room for them and the examiner is looking for whether the gap was noticed.
- 75 Ba’s “All of its missiles in XYZ were sold by Donald” is the same sentence with a typo; treat it identically.

2081 Bhadra and 2082 Bhadra: the two renamings that add a premise 9 6

81 Bh 9, 82 Bh 6

QUESTION AS PRINTED 2081 Bhadra · CT 710 · [9]

Every American who sells weapons to hostile nations is a criminal. Every enemy of America is a hostile. Nono has some missiles. All missiles of Nono were sold by Gorge. Gorge is an American. Nono is a Country. Nono is the enemy of America. Missiles are weapons. Prove that “Is Gorge a criminal?” using Resolution by Refutation.

QUESTION AS PRINTED 2082 Bhadra · CT 710 · [6]

It is a crime for a Nepali to sell weapons to an enemy of Nepal. Mr. A is an enemy of Nepal. Mr. A has some missiles. Missiles are weapons. Mr. R is a Nepali. If Mr. A has missiles then those missiles were sold to Mr. A by Mr. R. Prove “Mr. A is a criminal”.

2081 Bhadra: complete, with one spare premise

- All eight premises of the canonical set are present, plus “*Nono is a Country*”, *Country(Nono)*, which **appears in no rule and is never used**.
- Substitute [*Gorge/West*] and the eight steps run unchanged. Answer: **Gorge is a criminal.** ✓
- The question is phrased as “*Is Gorge a criminal?*” but asks for resolution *by refutation*, so still negate: assume $\neg \text{Criminal}(\text{Gorge})$ and derive $\square$. **Do not switch to extraction** just because the sentence ends in a question mark.

2082 Bhadra: the goal is the wrong person

- Read the roles: **Mr. R** is the Nepali who *sells*; **Mr. A** is the enemy who *buys*. The crime rule makes the *seller* the criminal.
- So the premises prove $\text{Criminal}(\text{Mr. R})$, and the printed goal “*Mr. A is a criminal*” does not follow.
- **How to answer:** prove $\text{Criminal}(\text{Mr. R})$ in full by the standard eight steps, then one line – “*the premises make the seller the criminal, so what follows is $\text{Criminal}(\text{Mr. R})$; $\text{Criminal}(\text{Mr. A})$ is not derivable from this set.*”
- The sixth premise is a conditional, *if Mr. A has missiles then they were sold to him by Mr. R*, which is the same clause 3 as ever:

$$\neg \text{Missile}(x) \vee \neg \text{Owns}(A, x) \vee \text{Sells}(R, x, A).$$
- *Enemy of Nepal* is used directly as the hostile predicate here, so no bridging axiom is needed.

P6. The same premises by forward and backward chaining

Prove Col. West is criminal using forward chaining 2+6

PIN 1/41 81 Ash 2+6 · the only paper that asks for a chaining proof rather than resolution, and 11 more ask for the forward/backward comparison this answers by example

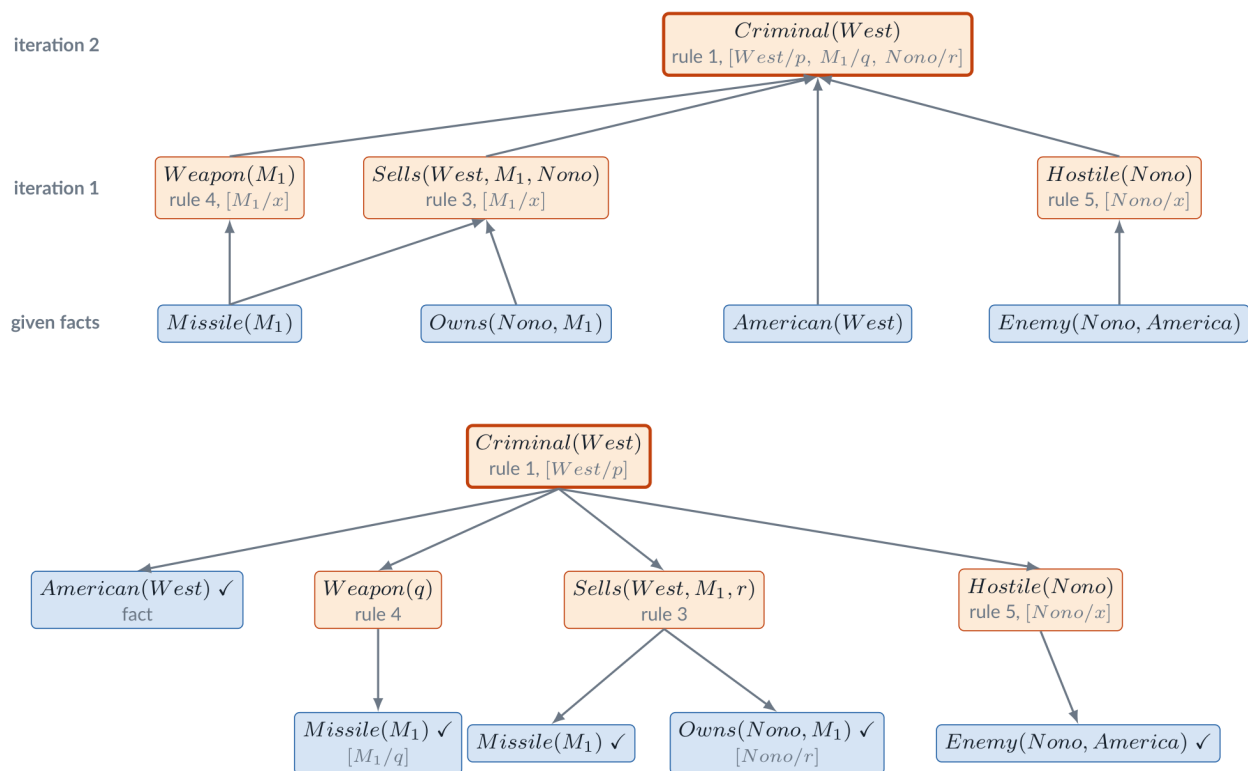
QUESTION AS PRINTED 2081 Ashwin · [2+6]

The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American. Prove that Col. West is criminal using **forward chaining**.

Forward chaining, iteration by iteration

- **Start with the ground facts** (the clauses with no premises):
 $American(West)$, $Missile(M_1)$, $Owns(Nono, M_1)$,
 $Enemy(Nono, America)$
- **Iteration 1** – fire every rule whose premises are all present:
 - rule 4 with $[M_1/x] \rightarrow Weapon(M_1)$
 - rule 3 with $[M_1/x]$, needing $Missile(M_1)$ and $Owns(Nono, M_1)$, both present $\rightarrow Sells(West, M_1, Nono)$
 - rule 5 with $[Nono/x] \rightarrow Hostile(Nono)$
 - rule 1 does **not** fire yet: its four premises are not all known.
- **Iteration 2** – rule 1 now matches with $[West/p, M_1/q, Nono/r]$, because $American(West)$, $Weapon(M_1)$, $Sells(West, M_1, Nono)$ and $Hostile(Nono)$ are all in the fact base:
 $\rightarrow Criminal(West)$
- **Goal reached, stop.** ✓

Draw it as three rows: the four given facts along the bottom, the three iteration-1 facts above them, $Criminal(West)$ at the top, with an arrow from each premise into the fact it helped derive. That picture is the answer; see §4.6.



The same thing backwards, for the 2 marks of comparison

- **Goal $Criminal(West)$.** Only rule 1 concludes $Criminal$, so unify $[West/p]$ and take its four premises as sub-goals:
 - $American(West)$ ✓ a fact, done;
 - $Weapon(q) \rightarrow$ sub-goal $Missile(q)$ ✓ matches the fact $Missile(M_1)$, binding $[M_1/q]$;
 - $Sells(West, M_1, r) \rightarrow$ sub-goals $Missile(M_1)$ ✓ and $Owns(Nono, M_1)$ ✓, binding $[Nono/r]$;
 - $Hostile(Nono) \rightarrow$ sub-goal $Enemy(Nono, America)$ ✓.
- Every leaf is a fact, so the goal is proved. **Four sub-goals, depth first, left to right.**

What the examiner is comparing

- Forward derived **three facts nobody asked for** on the way ($Weapon$, $Sells$, $Hostile$ are all needed here, but on a bigger KB most would not be).
- Backward touched **only the four conjuncts of one rule**, and the binding $[M_1/q]$ found in the second sub-goal was **carried into the third**, which is why $Sells(West, M_1, r)$ could be matched at all.
- Same rules, same modus ponens, opposite ends. One sentence, and the comparison marks are secured.

P5. Resolution refutation Family D: the light sleeper

2 papers, 17 marks 76 Ba **2+8**, 76 Bh **7**. The hardest set in the chapter, because the goal is itself an implication and the premises hide two nested quantifiers.

QUESTION AS PRINTED 2076 Baishakh · [2+8]

List down the rule for Inference. Consider the following axioms: All hounds howl at night. Anyone who has any cats will not have any mice. Light sleepers do not have anything which howls at night. John has either a cat or a hound. Prove: "If John is a light sleeper, then John does not have any mice" by resolution refutation.

Step 1 Facts into FOPL

1. $\forall x \text{ Hound}(x) \rightarrow \text{Howl}(x)$
 2. $\forall x \forall y [\text{Have}(x, y) \wedge \text{Cat}(y)] \rightarrow \forall z [\text{Have}(x, z) \rightarrow \neg \text{Mouse}(z)]$
 3. $\forall x \text{ LS}(x) \rightarrow \forall y [\text{Have}(x, y) \rightarrow \neg \text{Howl}(y)]$
 4. $\exists x \text{ Have}(\text{John}, x) \wedge [\text{Cat}(x) \vee \text{Hound}(x)]$
- Goal** $\text{LS}(\text{John}) \rightarrow \forall z [\text{Have}(\text{John}, z) \rightarrow \neg \text{Mouse}(z)]$

Step 2 Into CNF

1. $\neg \text{Hound}(x) \vee \text{Howl}(x)$
2. $\neg \text{Have}(x, y) \vee \neg \text{Cat}(y) \vee \neg \text{Have}(x, z) \vee \neg \text{Mouse}(z)$
3. $\neg \text{LS}(x) \vee \neg \text{Have}(x, y) \vee \neg \text{Howl}(y)$
- 4a. $\text{Have}(\text{John}, a)$ skolem constant a
- 4b. $\text{Cat}(a) \vee \text{Hound}(a)$

Step 3 Negate the goal, and get three clauses

- $\neg [\text{LS}(\text{John}) \rightarrow \forall z (\text{Have}(\text{John}, z) \rightarrow \neg \text{Mouse}(z))]$
 - $= \text{LS}(\text{John}) \wedge \exists z [\text{Have}(\text{John}, z) \wedge \text{Mouse}(z)]$
- 5a. $\text{LS}(\text{John})$ 5b. $\text{Have}(\text{John}, b)$ 5c. $\text{Mouse}(b)$
skolem b

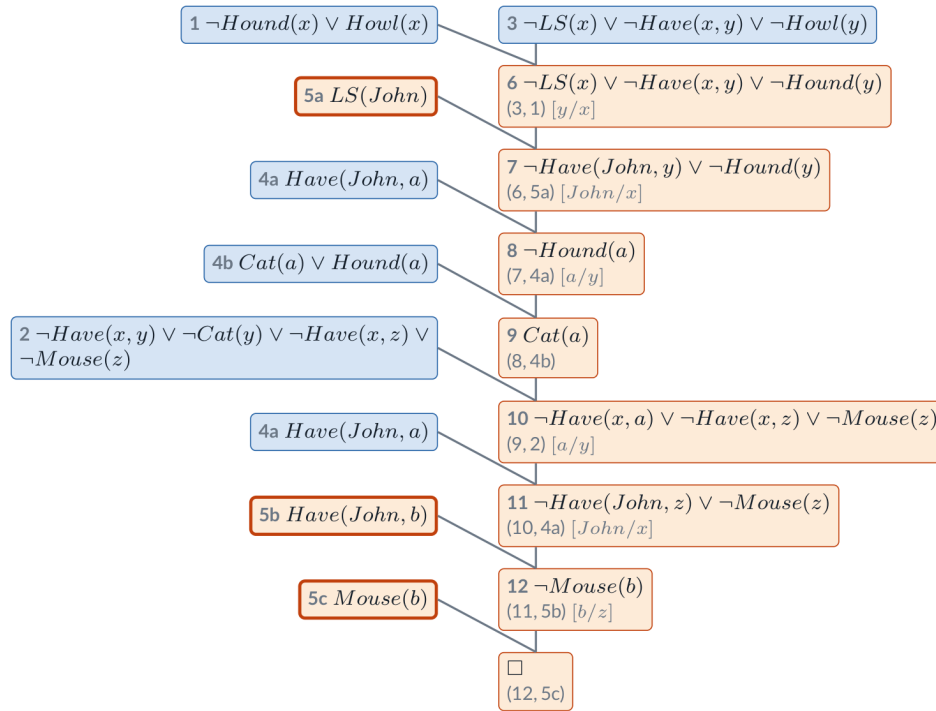
$\neg(P \rightarrow Q) = P \wedge \neg Q$. **This is the whole difficulty of the problem:** students negate an implication to another implication, get one clause instead of three, and then cannot close the proof.

Step 4 Resolution

6. $\neg \text{LS}(x) \vee \neg \text{Have}(x, y) \vee \neg \text{Hound}(y)$ (3, 1) $[y/x]$
7. $\neg \text{Have}(\text{John}, y) \vee \neg \text{Hound}(y)$ (6, 5a) $[\text{John}/x]$
8. $\neg \text{Hound}(a)$ (7, 4a) $[a/y]$
9. $\text{Cat}(a)$ (8, 4b)
10. $\neg \text{Have}(x, a) \vee \neg \text{Have}(x, z) \vee \neg \text{Mouse}(z)$ (9, 2) $[a/y]$
11. $\neg \text{Have}(\text{John}, z) \vee \neg \text{Mouse}(z)$ (10, 4a) $[\text{John}/x]$
12. $\neg \text{Mouse}(b)$ (11, 5b) $[b/z]$
13. $\square$ (12, 5c)

Final answer

- **"If John is a light sleeper then John has no mice" is proved.** ✓ Eight steps.
- **The shape of the argument**, worth stating in words underneath: John is a light sleeper, so he has nothing that howls, so what he has is not a hound; it was a cat or a hound, so it is a cat; anyone with a cat has no mice; so he has no mice.
- **Two skolem constants, and they are different.** a is the animal John owns (from premise 4); b is the mouse the negated goal claims he has. Reusing one letter for both is a wrong proof that happens to close.
- **Premise 2 needs four literals.** Both the cat and the mouse are things *John has*, so *Have* appears twice with different second arguments. Writing it with one *Have* loses the proof.



QUESTION AS PRINTED 2076 Bhadra · CT 78506 · [7]

Consider the following axioms: All direwolves howl at night. Anyone who has any cat will not have any mice. Light sleepers do not have anything that howls at night. Jon has either a cat or a direwolf. Prove that if Jon is a light sleeper, then Jon does not have any mice, using the resolution method.

Identical: substitute $[\text{Jon}/\text{John}, \text{Direwolf}/\text{Hound}]$ and the same eight steps close the proof. Nothing else changes.

P5. Resolution refutation The ten one-off premise sets

11 papers, one set each except Santa which is set twice 72 Ma 7, 81 Ch 8, 78 Ch 3+3, 80 Ch 3+5, 81 Ba 2+6, 79 Ash 7, 77 Ch 8, 70 Bh 8, 71 Bh 8, 73 Ma 10, 80 Ash 3+5. Same five steps every time; only the premises move.

1. Did Marcus hate Caesar? 7

72 Ma 7 · the paper prints "Find, did Marcus Caesar?", dropping the verb; it means *hate*

QUESTION AS PRINTED 2072 Magh · [7]

Using resolution, solve the following statements: All pompeian are Romans. All Romans were either loyal to Caesar or hated him. Everyone is loyal to someone. People only try to assassinate rulers they not loyal to. Marcus tried to assassinate Caesar. Marcus was pompeian. Find, did Marcus Caesar?

FOPL

1. $\forall x \text{Pompeian}(x) \rightarrow \text{Roman}(x)$
2. $\forall x \text{Roman}(x) \rightarrow \text{loyal}(x, \text{Caesar}) \vee \text{hate}(x, \text{Caesar})$
3. $\forall x \exists y \text{loyal}(x, y)$
4. $\forall x \forall y \text{tryAssassinate}(x, y) \wedge \text{ruler}(y) \rightarrow \neg \text{loyal}(x, y)$
5. $\text{tryAssassinate}(\text{Marcus}, \text{Caesar})$
6. $\text{Pompeian}(\text{Marcus})$ 7. $\text{ruler}(\text{Caesar})$ assumed
- Goal: $\text{hate}(\text{Marcus}, \text{Caesar})$

CNF

1. $\neg \text{Pompeian}(x) \vee \text{Roman}(x)$
2. $\neg \text{Roman}(x) \vee \text{loyal}(x, \text{Caesar}) \vee \text{hate}(x, \text{Caesar})$
3. $\text{loyal}(x, f(x))$ $\exists$ inside $\forall$: skolem function
4. $\neg \text{tryAssassinate}(x, y) \vee \neg \text{ruler}(y) \vee \neg \text{loyal}(x, y)$
5. $\text{tryAssassinate}(\text{Marcus}, \text{Caesar})$
6. $\text{Pompeian}(\text{Marcus})$ 7. $\text{ruler}(\text{Caesar})$

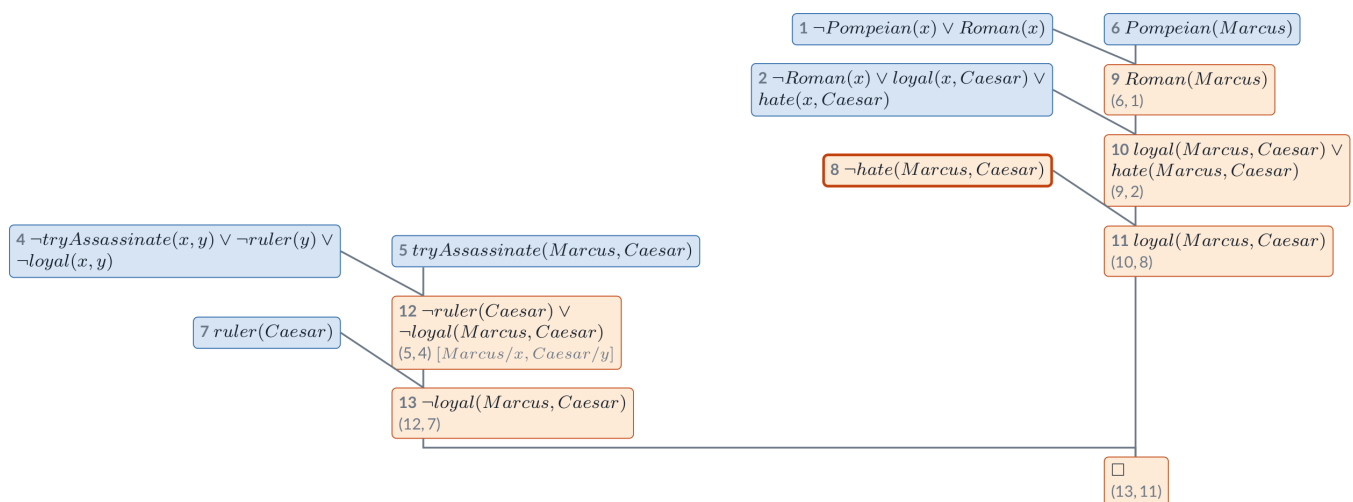
$\text{ruler}(\text{Caesar})$ is not printed but rule 4 is about rulers, so state it as an assumption. “Everyone is loyal to someone” is $\forall x \exists y \text{loyal}(x, y)$, and **it skolemizes to $f(x)$, not to a constant** – each person is loyal to their own someone.

Proof negated goal 8. $\neg \text{hate}(\text{Marcus}, \text{Caesar})$

9. $\text{Roman}(\text{Marcus})$ (6, 1)
10. $\text{loyal}(\text{Marcus}, \text{Caesar}) \vee \text{hate}(\text{Marcus}, \text{Caesar})$ (9, 2)
11. $\text{loyal}(\text{Marcus}, \text{Caesar})$ (10, 8)
12. $\neg \text{ruler}(\text{Caesar}) \vee \neg \text{loyal}(\text{Marcus}, \text{Caesar})$ (5, 4)
[$\text{Marcus}/x, \text{Caesar}/y$]
13. $\neg \text{loyal}(\text{Marcus}, \text{Caesar})$ (12, 7)
14. $\square$ (13, 11)

– **Marcus hated Caesar.** ✓

– **Clause 3 is never used**, and it is the only clause with a Skolem function in it. It is there to see whether you skolemize it correctly, not because the proof needs it.

**2. Scrooge is not a child** 8 3+3

81 Ch 8, 78 Ch 3+3 · 78 Ch prints the same six axioms and asks only for the CNF plus the horn-clause definition, so the clause list below is the whole answer there

QUESTION AS PRINTED 2081 Chaitra · [8]

Every child loves Santa. Everyone who loves Santa loves any reindeer. Rudolph is a reindeer, and Rudolph has a red nose. Anything which has a red nose is weird or is a clown. No reindeer is a clown. Scrooge does not love anything which is weird. Using Resolution Refutation, derive the conclusion: Scrooge is not a child.

FOPL

1. $\forall x \text{ child}(x) \rightarrow \text{loves}(x, \text{Santa})$
2. $\forall x \forall y \text{ loves}(x, \text{Santa}) \wedge \text{reindeer}(y) \rightarrow \text{loves}(x, y)$
3. $\text{reindeer}(\text{Rudolph})$ 4. $\text{redNose}(\text{Rudolph})$
5. $\forall x \text{ redNose}(x) \rightarrow \text{weird}(x) \vee \text{clown}(x)$
6. $\forall x \text{ reindeer}(x) \rightarrow \neg \text{clown}(x)$
7. $\forall x \text{ weird}(x) \rightarrow \neg \text{loves}(\text{Scrooge}, x)$

Goal: $\neg \text{child}(\text{Scrooge})$

"No reindeer is a clown" is $\neg \exists x (\text{reindeer}(x) \wedge \text{clown}(x))$; pushing the $\neg$ in gives line 6.

CNF

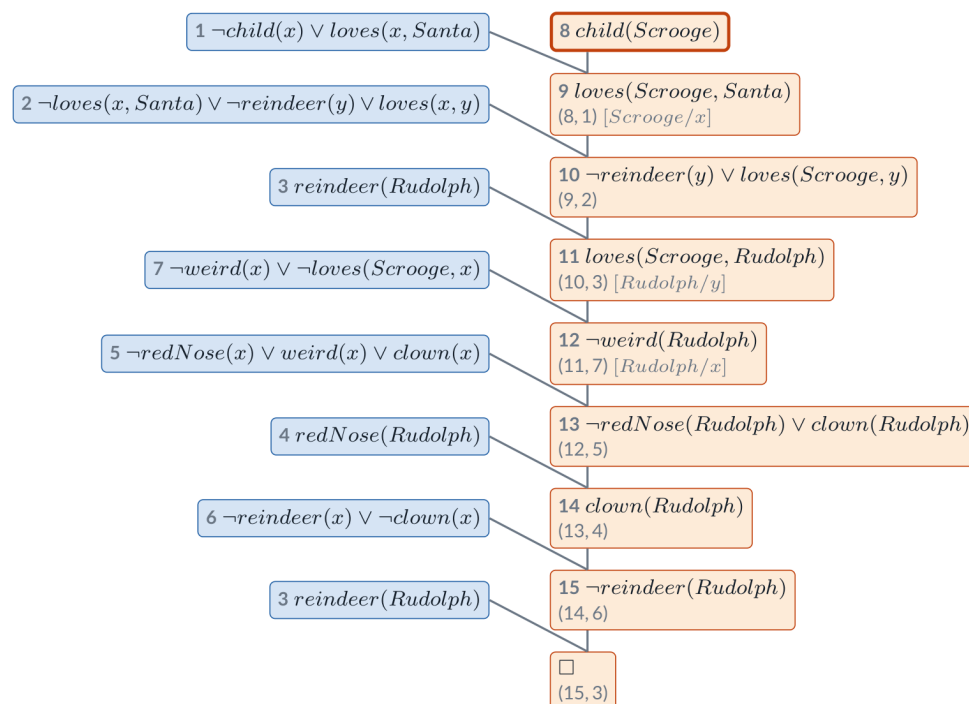
1. $\neg \text{child}(x) \vee \text{loves}(x, \text{Santa})$
2. $\neg \text{loves}(x, \text{Santa}) \vee \neg \text{reindeer}(y) \vee \text{loves}(x, y)$
3. $\text{reindeer}(\text{Rudolph})$ 4. $\text{redNose}(\text{Rudolph})$
5. $\neg \text{redNose}(x) \vee \text{weird}(x) \vee \text{clown}(x)$
6. $\neg \text{reindeer}(x) \vee \neg \text{clown}(x)$
7. $\neg \text{weird}(x) \vee \neg \text{loves}(\text{Scrooge}, x)$

The goal is already a negation, so **negating it removes the $\neg$** : clause 8 is $\text{child}(\text{Scrooge})$, not $\neg \text{child}(\text{Scrooge})$. Getting this backwards is the standard error on this set.

Proof **negated goal** 8. $\text{child}(\text{Scrooge})$

9. $\text{loves}(\text{Scrooge}, \text{Santa})$ (8, 1) [$\text{Scrooge}/x$]
10. $\neg \text{reindeer}(y) \vee \text{loves}(\text{Scrooge}, y)$ (9, 2)
11. $\text{loves}(\text{Scrooge}, \text{Rudolph})$ (10, 3) [$\text{Rudolph}/y$]
12. $\neg \text{weird}(\text{Rudolph})$ (11, 7) [$\text{Rudolph}/x$]
13. $\neg \text{redNose}(\text{Rudolph}) \vee \text{clown}(\text{Rudolph})$ (12, 5)
14. $\text{clown}(\text{Rudolph})$ (13, 4)
15. $\neg \text{reindeer}(\text{Rudolph})$ (14, 6)
16. $\square$ (15, 3)

- **"Scrooge is not a child" is proved.** ✓ Eight steps, every premise used.
- The argument in words: if Scrooge were a child he would love Santa, hence love Rudolph; but Scrooge loves nothing weird, so Rudolph is not weird, so (red nose) he is a clown, and no reindeer is a clown – contradiction.

**3. Did Curiosity kill the cat?** 3+5

PIN 1/41 80 Ch 3+5 · **the hardest clause set in the chapter:** premise (a) is the Skolem-function sentence from P2, and it is needed twice

QUESTION AS PRINTED 2080 Chaitra · [3+5]

Convert the following sentences into FOPL and answer "Did Curiosity kill the cat" by resolution refutation:

- a) Everyone who loves all animals is loved by someone.
- b) Anyone who kills an animal is loved by no one.
- c) Jack loves all animals.
- d) Either Jack or Curiosity killed the cat.

FOPL (the 3 marks)

- A. $\forall x [\forall y \text{ Animal}(y) \rightarrow \text{Loves}(x, y)] \rightarrow \exists z \text{ Loves}(z, x)$
 B. $\forall x [\exists y \text{ Animal}(y) \wedge \text{Kills}(x, y)] \rightarrow \forall z \neg \text{Loves}(z, x)$
 C. $\forall x \text{ Animal}(x) \rightarrow \text{Loves}(Jack, x)$
 D. $\text{Kills}(Jack, Tuna) \vee \text{Kills}(Curiosity, Tuna)$
 E. $\text{Cat}(Tuna)$ F. $\forall x \text{ Cat}(x) \rightarrow \text{Animal}(x)$ assumed
Goal: $\text{Kills}(Curiosity, Tuna)$

CNF (the cat is named Tuna; "cats are animals" is assumed)

- A1. $\text{Animal}(F(x)) \vee \text{Loves}(G(x), x)$
 A2. $\neg \text{Loves}(x, F(x)) \vee \text{Loves}(G(x), x)$
 B. $\neg \text{Animal}(y) \vee \neg \text{Kills}(x, y) \vee \neg \text{Loves}(z, x)$
 C. $\neg \text{Animal}(x) \vee \text{Loves}(Jack, x)$
 D. $\text{Kills}(Jack, Tuna) \vee \text{Kills}(Curiosity, Tuna)$
 E. $\text{Cat}(Tuna)$ F. $\neg \text{Cat}(x) \vee \text{Animal}(x)$

A1 and A2 are premise (a) put through all eight CNF steps in P2 above. $F(x)$ is "an animal x does not love" and $G(x)$ is "someone who loves x ". **Both must be functions of x .**

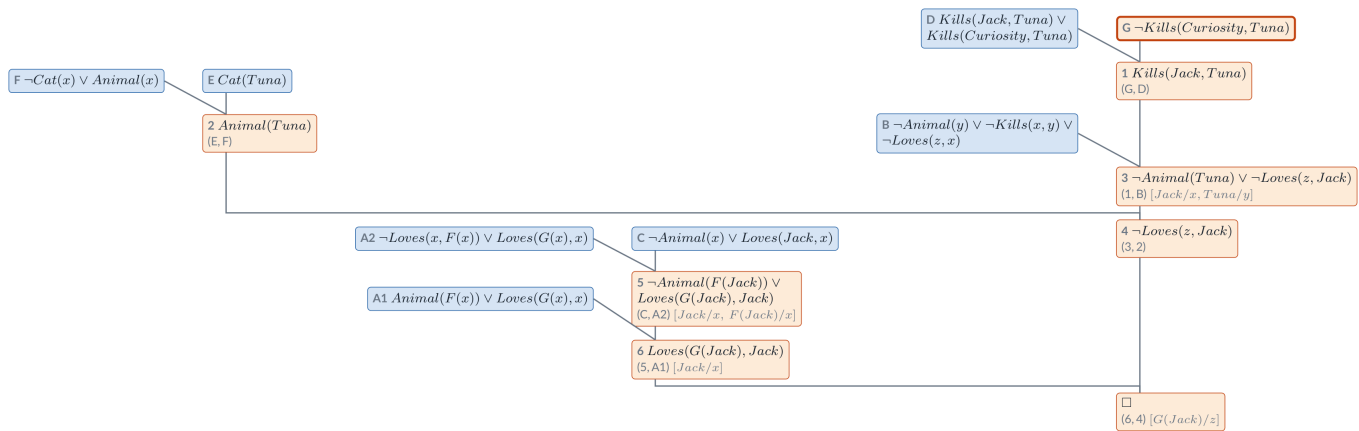
Proof negated goal G. $\neg \text{Kills}(Curiosity, Tuna)$

1. $\text{Kills}(Jack, Tuna)$ (G, D)
2. $\text{Animal}(Tuna)$ (E, F)
3. $\neg \text{Animal}(Tuna) \vee \neg \text{Loves}(z, Jack)$ (1, B)
 $[Jack/x, Tuna/y]$
4. $\neg \text{Loves}(z, Jack)$ (3, 2)
5. $\neg \text{Animal}(F(Jack)) \vee \text{Loves}(G(Jack), Jack)$ (C, A2)
 $[Jack/x, F(Jack)/x]$
6. $\text{Loves}(G(Jack), Jack)$ (5, A1) $[Jack/x]$, duplicate collapses
7. $\square$ (6, 4) $[G(Jack)/z]$

– **Curiosity killed the cat.** ✓

– **Steps 5 and 6 are the whole difficulty.** They prove somebody loves Jack, using (c) to satisfy the antecedent of (a): Jack loves all animals, so in particular he loves $F(Jack)$, so the second clause of (a) fires. Only then does step 4, *nobody loves Jack*, become a contradiction.

– **Two facts have to be assumed** because the paper omits them: the cat has a name, and cats are animals. State both.

**4. Sneha is happy** 2+6

81 Ba 2+6 · the first 2 marks are skolemization defined, §4.3 · **contrast the CNF of premise (b) with the food family's premise 4:** here the distribution into two clauses is valid

QUESTION AS PRINTED 2081 Baishakh · CT 710 · [2+6]

Define skolemization with example. Consider the following axioms: a) anyone passing his engineering exams and winning the lottery is happy; b) anyone who studies or is lucky can pass all his exams; c) Sneha did not study but she is lucky; d) anyone who is lucky wins the lottery. Using resolution refutation, prove that Sneha is happy.

FOPL

1. $\forall x \text{ pass}(x) \wedge \text{win}(x) \rightarrow \text{happy}(x)$
2. $\forall x \text{ study}(x) \vee \text{lucky}(x) \rightarrow \text{pass}(x)$
3. $\neg \text{study}(\text{Sneha}) \wedge \text{lucky}(\text{Sneha})$
4. $\forall x \text{ lucky}(x) \rightarrow \text{win}(x)$

Goal: $\text{happy}(\text{Sneha})$

CNF

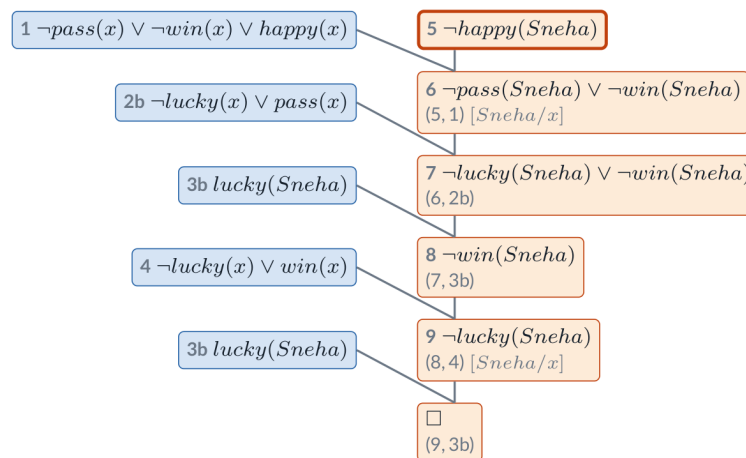
1. $\neg \text{pass}(x) \vee \neg \text{win}(x) \vee \text{happy}(x)$
2. $[\text{study}(x) \vee \text{lucky}(x)] \rightarrow \text{pass}(x)$
 $= [\neg \text{study}(x) \wedge \neg \text{lucky}(x)] \vee \text{pass}(x)$, so
 - 2a. $\neg \text{study}(x) \vee \text{pass}(x)$
 - 2b. $\neg \text{lucky}(x) \vee \text{pass}(x)$
- 3a. $\neg \text{study}(\text{Sneha})$ **3b.** $\text{lucky}(\text{Sneha})$
4. $\neg \text{lucky}(x) \vee \text{win}(x)$

Why this splits and the food premise does not: here the negation lands on a *disjunction*, $\neg(A \vee B) = \neg A \wedge \neg B$, so distributing is forced. In the food family it lands on a *conjunction*, $\neg(A \wedge \neg B) = \neg A \vee B$, and there is nothing to split. One De Morgan apart, opposite answers.

Proof negated goal 5. $\neg \text{happy}(\text{Sneha})$

6. $\neg \text{pass}(\text{Sneha}) \vee \neg \text{win}(\text{Sneha})$ (5, 1) $[\text{Sneha}/x]$
7. $\neg \text{lucky}(\text{Sneha}) \vee \neg \text{win}(\text{Sneha})$ (6, 2b)
8. $\neg \text{win}(\text{Sneha})$ (7, 3b)
9. $\neg \text{lucky}(\text{Sneha})$ (8, 4) $[\text{Sneha}/x]$
10. $\square$ (9, 3b)

- **“Sneha is happy” is proved.** ✓ Five steps.
- **Clauses 2a and 3a are never used.** “Sneha did not study” is there only to tempt you into deriving nothing from it; the lucky half carries the whole proof.
- “can pass all his exams” is written $\text{pass}(x)$ with the exam left implicit. Adding a second argument $\text{pass}(x, \text{exam})$ changes nothing except the typing.

**5. Steve likes BK301** 7

79 Ash 7 · **the printed premise runs the wrong way** and has to be read as its converse before anything can be proved

QUESTION AS PRINTED 2079 Ashwin · [7]

Assume the following facts: a) Steve only likes easy courses; b) Science courses are hard; c) All the courses in the basket weaving department are easy; d) BK301 is a basket weaving course. Prove that Steve likes BK301 using Resolution Refutation.

Read premise (a) carefully first

- Literally, “Steve only likes easy courses” is $\forall x \text{ likes}(\text{Steve}, x) \rightarrow \text{easy}(x)$: liking implies easy.
- From that you can prove a course Steve likes is easy. You cannot prove he likes anything, because nothing in the set ever concludes *likes*.
- The question is only answerable under the converse reading, “Steve likes every easy course”:

$$\forall x \text{ easy}(x) \rightarrow \text{likes}(\text{Steve}, x)$$

- **Write that line, then prove it.** Stating the reading you adopted is a mark; silently reversing an implication is not.

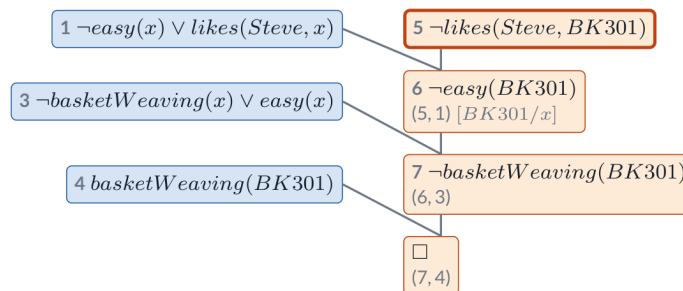
FOPL

1. $\forall x \text{ easy}(x) \rightarrow \text{likes}(\text{Steve}, x)$ converse reading
 2. $\forall x \text{ science}(x) \rightarrow \neg \text{easy}(x)$
 3. $\forall x \text{ basketWeaving}(x) \rightarrow \text{easy}(x)$
 4. $\text{basketWeaving}(\text{BK301})$
- Goal:** $\text{likes}(\text{Steve}, \text{BK301})$

CNF and proof

1. $\neg \text{easy}(x) \vee \text{likes}(\text{Steve}, x)$ (a), converse reading
2. $\neg \text{science}(x) \vee \neg \text{easy}(x)$ (b)
3. $\neg \text{basketWeaving}(x) \vee \text{easy}(x)$ (c)
4. $\text{basketWeaving}(\text{BK301})$ (d)
5. $\neg \text{likes}(\text{Steve}, \text{BK301})$ negated goal
6. $\neg \text{easy}(\text{BK301})$ (5, 1) [BK301/x]
7. $\neg \text{basketWeaving}(\text{BK301})$ (6, 3)
8. $\square$ (7, 4)

- **“Steve likes BK301” is proved** under that reading. ✓ Three steps.
- **Premise (b) is unused**, and it is the only one mentioning science. *Hard* is written $\neg \text{easy}$; inventing a separate *hard* predicate needs a fifth premise saying the two are opposites.

**6. You will enjoy 8**

77 Ch 8 · purely propositional, no quantifiers, and **four of the seven premises are noise**

QUESTION AS PRINTED 2077 Chaitra · CT 78506 · [8]

If it is a sunny and warm day you will enjoy. If it is a warm and pleasant day you will do strawberry picking. If it is raining, no strawberry picking. If it is raining you will get wet. It is a warm day. It is raining. It is sunny. Prove that “You will enjoy” using resolution by refutation.

Into logic (propositional: no objects, so no quantifiers)

1. $\text{sunny} \wedge \text{warm} \rightarrow \text{enjoy}$
 2. $\text{warm} \wedge \text{pleasant} \rightarrow \text{picking}$
 3. $\text{raining} \rightarrow \neg \text{picking}$
 4. $\text{raining} \rightarrow \text{wet}$
 5. warm 6. raining 7. sunny
- Goal:** enjoy

CNF

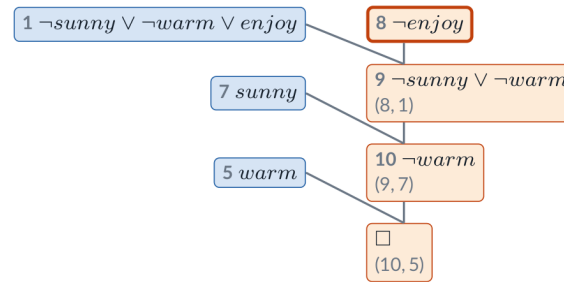
1. $\neg \text{sunny} \vee \neg \text{warm} \vee \text{enjoy}$
2. $\neg \text{warm} \vee \neg \text{pleasant} \vee \text{picking}$
3. $\neg \text{raining} \vee \neg \text{picking}$
4. $\neg \text{raining} \vee \text{wet}$
5. warm 6. raining 7. sunny

Each implication becomes one clause by $\alpha \rightarrow \beta = \neg \alpha \vee \beta$ plus De Morgan on the conjunctive antecedent. No quantifiers means no unification: every resolution step here is on a **ground** literal.

Proof negated goal 8. $\neg \text{enjoy}$

9. $\neg \text{sunny} \vee \neg \text{warm}$ (8, 1)
10. $\neg \text{warm}$ (9, 7)
11. $\square$ (10, 5)

- **“You will enjoy” is proved** in three steps. ✓
- **Clauses 2, 3, 4 and 6 are never touched.** The strawberry picking and the rain are a complete decoy: they are consistent with the rest and irrelevant to the goal.
- **The lesson worth writing under the proof:** resolve *from the negated goal*. Starting from the facts, this set will happily generate *wet*, $\neg \text{picking}$ and several other true-but-useless literals before it stumbles on the answer.



7. Ram is naughty 8

70 Bh 8 · also set as an exercise by the textbook, whose own solution omits the argument order

QUESTION AS PRINTED 2070 Bhadra · [8]

All oversmart persons are stupid. Children of oversmart persons are naughty. Ram is children of Hari. Hari is oversmart. Show that Ram is naughty using FOPL based resolution.

FOPL

1. $\forall x \text{ oversmart}(x) \rightarrow \text{stupid}(x)$
2. $\forall x \forall y \text{ child}(x, y) \wedge \text{oversmart}(y) \rightarrow \text{naughty}(x)$
3. $\text{child}(\text{Ram}, \text{Hari})$ 4. $\text{oversmart}(\text{Hari})$

Goal: $\text{naughty}(\text{Ram})$

CNF

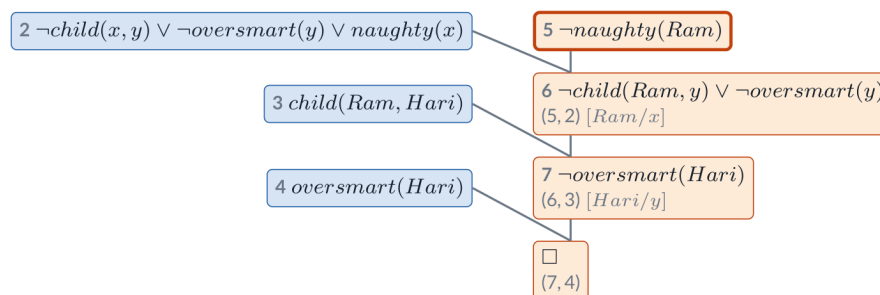
1. $\neg \text{oversmart}(x) \vee \text{stupid}(x)$
2. $\neg \text{child}(x, y) \vee \neg \text{oversmart}(y) \vee \text{naughty}(x)$
3. $\text{child}(\text{Ram}, \text{Hari})$ 4. $\text{oversmart}(\text{Hari})$

Proof negated goal 5. $\neg \text{naughty}(\text{Ram})$

6. $\neg \text{child}(\text{Ram}, y) \vee \neg \text{oversmart}(y)$ (5, 2) [Ram/x]
7. $\neg \text{oversmart}(\text{Hari})$ (6, 3) [Hari/y]
8. $\square$ (7, 4)

Notes

- “**Ram is naughty**” is proved in three steps. ✓
- **Premise 1 is unused.** *Stupid* appears nowhere else, so the whole first sentence is a decoy – the same trick as the mammal rule in Family B.
- $\text{child}(x, y)$ **must mean** “*x* is a child of *y*” and be used that way in both clause 2 and clause 3. The English “*Ram is children of Hari*” fixes the order; write the reading down before you write the clause, because reversing it silently breaks the unification in step 6.
- Predicate names are free, but *naughty* must be a *property of the child*, not a relation: $\text{naughty}(x)$, one argument.



8. The book supports the cup 8

71 Bh 8 · the only set written in **relational** rather than sentence form, so the argument order is the whole question

QUESTION AS PRINTED 2071 Bhadra · [8]

If X is on the top of Y, Y supports X. If X is above Y and they are touching each other, X is on the top of Y. A cup is above a book. A cup is touching a book. Show that $\text{supports}(\text{book}, \text{cup})$ is true.

FOPL

1. $\forall x \forall y \text{ onTop}(x, y) \rightarrow \text{supports}(y, x)$
2. $\forall x \forall y \text{ above}(x, y) \wedge \text{touching}(x, y) \rightarrow \text{onTop}(x, y)$
3. $\text{above}(\text{Cup}, \text{Book})$ 4. $\text{touching}(\text{Cup}, \text{Book})$

Goal: $\text{supports}(\text{Book}, \text{Cup})$

CNF

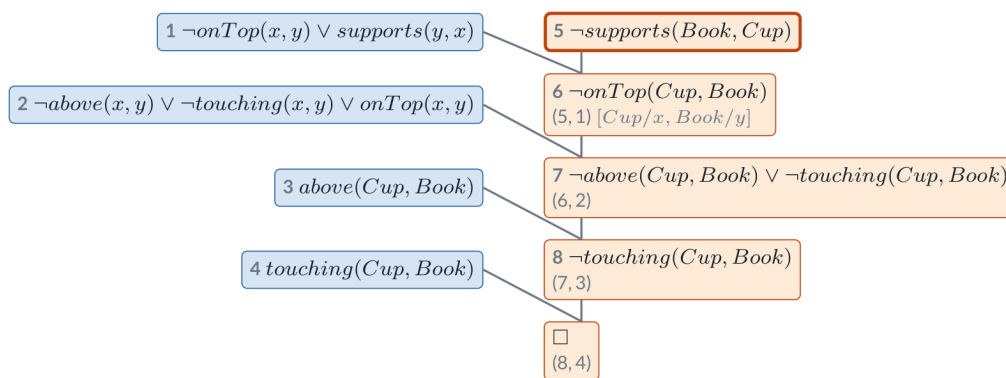
1. $\neg \text{onTop}(x, y) \vee \text{supports}(y, x)$
2. $\neg \text{above}(x, y) \vee \neg \text{touching}(x, y) \vee \text{onTop}(x, y)$
3. $\text{above}(\text{Cup}, \text{Book})$ 4. $\text{touching}(\text{Cup}, \text{Book})$

Proof negated goal 5. $\neg \text{supports}(\text{Book}, \text{Cup})$

6. $\neg \text{onTop}(\text{Cup}, \text{Book})$ (5, 1) $[\text{Cup}/x, \text{Book}/y]$
7. $\neg \text{above}(\text{Cup}, \text{Book}) \vee \neg \text{touching}(\text{Cup}, \text{Book})$ (6, 2)
8. $\neg \text{touching}(\text{Cup}, \text{Book})$ (7, 3)
9. $\square$ (8, 4)

Notes

- $\text{supports}(\text{Book}, \text{Cup})$ is proved in four steps. ✓
- **Clause 1 reverses its arguments and that is the point.** $\text{onTop}(x, y)$ has the supporter second; $\text{supports}(y, x)$ has it first. The goal $\text{supports}(\text{Book}, \text{Cup})$ therefore unifies with $[\text{Cup}/x, \text{Book}/y]$, giving $\text{onTop}(\text{Cup}, \text{Book})$ – cup on book, which is what the facts describe.
- **Every premise is used**, and there is no quantifier subtlety: the two rules are plain universals, so drop the $\forall$ and go.
- “they are touching each other” is symmetric in English but $\text{touching}(x, y)$ is not symmetric in the clause set. It does not matter here because the facts are given in the same order the rule wants; if they were not, a symmetry axiom $\neg \text{touching}(x, y) \vee \text{touching}(y, x)$ would be needed.



9. If John does not study, Mary does not love John 10

PIN 1/41 73 Ma 10 · the goal is an implication again, so **negating it gives two clauses**, and every premise is a contrapositive

QUESTION AS PRINTED 2073 Magh · [10]

Consider the following axioms: i) anyone whom Mary loves is a football star; ii) any student who does not pass does not play; iii) John is a student; iv) any student who does not study does not pass; v) anyone who does not play is not a football star. Prove that “If John does not study, then Mary does not love John” using resolution by refutation.

FOPL

1. $\forall x \text{ loves}(\text{Mary}, x) \rightarrow \text{star}(x)$
 2. $\forall x \text{ student}(x) \wedge \neg \text{pass}(x) \rightarrow \neg \text{play}(x)$
 3. $\text{student}(\text{John})$
 4. $\forall x \text{ student}(x) \wedge \neg \text{study}(x) \rightarrow \neg \text{pass}(x)$
 5. $\forall x \neg \text{play}(x) \rightarrow \neg \text{star}(x)$
- Goal:** $\neg \text{study}(\text{John}) \rightarrow \neg \text{loves}(\text{Mary}, \text{John})$

CNF

1. $\neg \text{loves}(\text{Mary}, x) \vee \text{star}(x)$
2. $\neg \text{student}(x) \vee \text{pass}(x) \vee \neg \text{play}(x)$
3. $\text{student}(\text{John})$
4. $\neg \text{student}(x) \vee \text{study}(x) \vee \neg \text{pass}(x)$
5. $\text{play}(x) \vee \neg \text{star}(x)$

Negate the goal: two clauses

- $\neg[\neg \text{study}(\text{John}) \rightarrow \neg \text{loves}(\text{Mary}, \text{John})] =$
 $\neg \text{study}(\text{John}) \wedge \text{loves}(\text{Mary}, \text{John})$

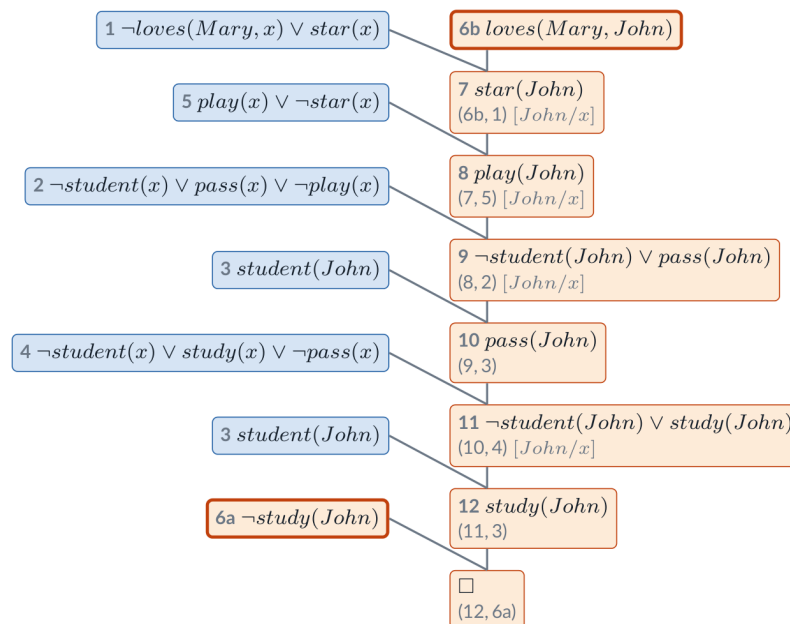
6a. $\neg \text{study}(\text{John})$ 6b. $\text{loves}(\text{Mary}, \text{John})$

Premises (ii), (iv) and (v) are all of the form *not-P implies not-Q*. Convert each one mechanically: $[S \wedge \neg P] \rightarrow \neg Q$ becomes $\neg S \vee P \vee \neg Q$. Trying to simplify them into positive form first is where this question is usually lost.

Proof

7. $\text{star}(\text{John})$ (6b, 1) [John/x]
8. $\text{play}(\text{John})$ (7, 5) [John/x]
9. $\neg \text{student}(\text{John}) \vee \text{pass}(\text{John})$ (8, 2) [John/x]
10. $\text{pass}(\text{John})$ (9, 3)
11. $\neg \text{student}(\text{John}) \vee \text{study}(\text{John})$ (10, 4) [John/x]
12. $\text{study}(\text{John})$ (11, 3)
13. $\square$ (12, 6a)

- **The implication is proved.** ✓ Seven steps, every premise used, and clause 3 used **twice**.
- The chain in words: suppose Mary loves John and he does not study. Loved by Mary $\rightarrow$ football star $\rightarrow$ plays $\rightarrow$ (being a student) passes $\rightarrow$ studies. Contradiction with *does not study*.
- **A clause may be reused as often as needed.** Students often cross a clause off after one use; nothing in resolution says that.

**10. Can anyone be found with an exciting life? 3+5**

80 Ash 3+5 · asks *who*, not *prove*, so this is the chapter's one genuine **answer-extraction** question

QUESTION AS PRINTED 2080 Ashwin · [3+5]

List the rules of inference. Convert the following into FOPL and prove using Resolution Refutation System: "All people who are not poor and are smart are happy. Those people who can read are not stupid. John can read and is wealthy. Happy people have exciting lives. Can anyone be found with an exciting life?"

FOPL (the 5 marks begin here)

1. $\forall x \neg \text{poor}(x) \wedge \text{smart}(x) \rightarrow \text{happy}(x)$
2. $\forall x \text{read}(x) \rightarrow \neg \text{stupid}(x)$
3. $\text{read}(\text{John}) \wedge \text{wealthy}(\text{John})$, i.e.
 $\text{read}(\text{John}) \wedge \neg \text{poor}(\text{John})$
4. $\forall x \text{happy}(x) \rightarrow \text{exciting}(x)$
5. $\forall x \neg \text{stupid}(x) \rightarrow \text{smart}(x)$

bridge, stated

Goal: $\exists x \text{exciting}(x)$ **CNF** with $\text{wealthy} = \neg \text{poor}$ and $\text{smart} = \neg \text{stupid}$

1. $\text{poor}(x) \vee \neg \text{smart}(x) \vee \text{happy}(x)$
2. $\neg \text{read}(x) \vee \neg \text{stupid}(x)$
- 3a. $\text{read}(\text{John})$ 3b. $\neg \text{poor}(\text{John})$
4. $\neg \text{happy}(x) \vee \text{exciting}(x)$
5. $\neg \text{smart}(x) \vee \neg \text{stupid}(x)$ is **not** needed; instead state the bridging fact $\neg \text{stupid}(x) \rightarrow \text{smart}(x)$, i.e. 5.
 $\text{stupid}(x) \vee \text{smart}(x)$

Two vocabulary bridges must be stated: wealthy is $\neg \text{poor}$, and not stupid is smart . Without them the clause set has five predicates that never meet and no proof exists. Say so in one line before starting — the examiner's own wording forces it.

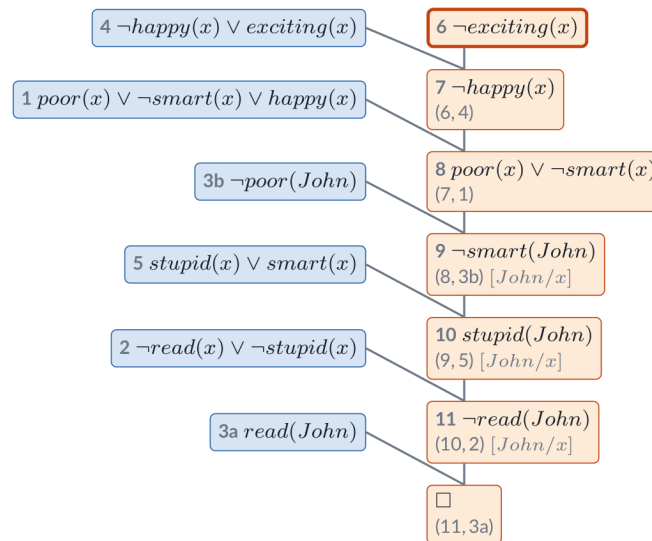
Proof negated goal 6. $\neg \text{exciting}(x)$, for all x

7. $\neg \text{happy}(x)$ (6, 4)
8. $\text{poor}(x) \vee \neg \text{smart}(x)$ (7, 1)
9. $\neg \text{smart}(\text{John})$ (8, 3b) [John/x]
10. $\text{stupid}(\text{John})$ (9, 5) [John/x]
11. $\neg \text{read}(\text{John})$ (10, 2) [John/x]
12. $\square$ (11, 3a)

– **Yes: John has an exciting life.** ✓

– **How the name is recovered**, which is what “can anyone be found” asks: the goal $\exists x \text{exciting}(x)$ negates to $\forall x \neg \text{exciting}(x)$, one clause with a free variable. The substitutions used along the way bound that variable to **John**, and that binding is the answer.

– The formal device is an **answer literal**: carry $\text{exciting}(x) \vee \text{Ans}(x)$ instead of the bare negation, and read the answer off the clause that survives. Worth naming even if you do not use it.

**P7. Bayes' theorem: the twelve numericals**

15 papers set a Bayes question; twelve give numbers and are worked below, three ask for a worked example of your own and are answered by any one of them. Every answer here was recomputed independently in `verify.py`, not copied from a source. The recipe never changes:

The recipe

1. **Name the events** and write down which probability each given number is. **Half the lost marks happen here**, in the first thirty seconds.
2. Write Bayes' theorem with the names, not with A and B .
3. If $P(\text{evidence})$ is not given, build it with the law of total probability: $P(E) = P(E|H)P(H) + P(E|\neg H)P(\neg H)$.
4. Substitute, and quote the answer as a percentage as well as a decimal.

B1–B2. Meningitis and the stiff neck 3+5 2+671 Ma 3+5, 81 Ba 2+6 · same scenario, **different numbers in each paper** — read them off the paper, not from memory**QUESTION AS PRINTED** 2071 Magh · [3+5]

What is neural network? Describe its types. A doctor knows that the disease meningitis causes the patient to have a stiff neck 50% of the time. The doctor also knows that the probability that a patient has meningitis is $1/50,000$, and the probability that any patient has a stiff neck is $1/20$. Find the probability that a patient with a stiff neck has meningitis?

B1 2071 Magh

- **Given** $P(s|m) = 0.5$, $P(m) = 1/50000 = 2 \times 10^{-5}$,
 $P(s) = 1/20 = 0.05$.
- **Wanted** $P(m|s)$.

$$P(m|s) = \frac{P(s|m)P(m)}{P(s)} = \frac{0.5 \times 2 \times 10^{-5}}{0.05} = \frac{10^{-5}}{0.05}$$

$$P(m|s) = 0.0002 = 0.02\%$$

- $P(s)$ is given here, so no total-probability step is needed. That is the only easy version of this question in the archive.

B2 2081 Baishakh (CT 710)

- **Given** $P(s|m) = 0.8$, $P(m) = 1/50000$, $P(s) = 0.01$.

$$P(m|s) = \frac{0.8 \times 2 \times 10^{-5}}{0.01} = \frac{1.6 \times 10^{-5}}{0.01}$$

$$P(m|s) = 0.0016 = 0.16\%$$

- The first 2 marks are **prior Vs posterior** defined: $P(m)$ is the prior, $P(m|s)$ the posterior, and the stiff neck raised the belief from **0.002 % to 0.16 %** — eighty times higher, and still under one in six hundred.
- **That last sentence is the exam answer to “what does this tell you”**: a strong likelihood cannot outweigh a tiny prior.

B3. Measles or flu? 8

73 Ma 8 · the priors are given directly as a 90/10 split, so this is the cleanest total-probability example in the archive

QUESTION AS PRINTED 2073 Magh · [8]

A doctor is called to see a sick child. The doctor has prior information that 90% of sick children in that neighborhood have the flu, while the other 10% are sick with measles. Let F stand for an event of a child being sick with Flu and M stand for an event of a child being sick with measles. Assume for simplicity that there no other maladies in that neighborhood. A well-known (and common) symptom of measles is a rash and has probability of 0.95. However, very occasionally, children with flu also develop rash and has probability of 0.08. Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?

Solution

- **Given** $P(F) = 0.9$, $P(M) = 0.1$, $P(R|M) = 0.95$,
 $P(R|F) = 0.08$.
- **Total probability of a rash:**

$$\begin{aligned} P(R) &= P(R|M)P(M) + P(R|F)P(F) \\ &= 0.95 \times 0.1 + 0.08 \times 0.9 = 0.095 + 0.072 = 0.167 \end{aligned}$$

$$P(M|R) = \frac{P(R|M)P(M)}{P(R)} = \frac{0.095}{0.167}$$

$$P(M|R) = 0.5689 \approx 56.9\%$$

Reading the answer

- Measles goes from a **10 % prior to a 57 % posterior**: the rash is strong evidence, but flu is nine times commoner, so it stays a coin flip.
- **Check the complement**:
 $P(F|R) = 0.072/0.167 = 0.4311$, and
 $0.5689 + 0.4311 = 1$. ✓ Two extra lines, and they catch an arithmetic slip in the denominator every time.
- **The trap**: reading 0.95 as $P(M|R)$ and answering “95 %”. The paper gives the *likelihood* $P(R|M)$; the whole point of the question is that the two are not the same number.

B4. The cigar smoker 2+6 880 Bh 2+6, 73 Bh 8 · two papers, identical data · part (i) is the **prior**, and it is one mark for writing down a number already in the question**QUESTION AS PRINTED** 2080 Bhadra · CT 710 · [2+6] also 2073 Bhadra CT 78506 [8]

What do you understand by Bayesian Network? In a County, 51% of the adults are males and the other 49% are females. One adult is randomly selected for a survey involving credit card usage. i) Find the prior probability that the selected person is a male. ii) It is later learned that the selected survey subject was smoking a cigar. Also, 9.5% of males smoke cigars, whereas 1.7% of females smoke cigars (based on data from the Substance Abuse and Mental Health Services Administration). Use this additional information to find the probability that the selected subject is a male.

Solution

- (i) **Prior** $P(M) = 0.51$. **That is the whole answer to part (i)**; no calculation is intended.
- (ii) **Given** $P(F) = 0.49$, $P(C|M) = 0.095$, $P(C|F) = 0.017$.

$$P(C) = 0.095 \times 0.51 + 0.017 \times 0.49 \\ = 0.04845 + 0.00833 = 0.05678$$

$$P(M|C) = \frac{0.04845}{0.05678}$$

$$P(M|C) = 0.8533 \approx 85.3\%$$

Notes

- The prior 51 % rises to **85 %** because men are five and a half times as likely to smoke a cigar. **Here the likelihood ratio does dominate**, which is the opposite lesson to B1 and B2 – and the difference is entirely that this prior is near a half rather than near zero.
- **Keep five decimal places in the denominator.** Rounding 0.05678 to 0.057 moves the answer to 85.0 %, and the marks for the final figure are awarded on the exact value.
- **73 Bh** prints the same scenario for a single 8-mark part. Same two answers.

B5. The 99 % accurate test for a rare disease 8

PIN 1/41 72 Ash 8 · **the base-rate question**, and the one every paper's theory part is really about

QUESTION AS PRINTED 2072 Ashwin · CT 78506 · [8]

After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e. the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given that you don't have the disease). The good news is that this is a rare disease, striking only one in 10,000 people. Why is it good news that the disease is rare? What are the chances that you actually have the disease?

Solution

- **Given** $P(D) = 10^{-4}$, $P(+|D) = 0.99$, $P(-|\neg D) = 0.99$, so $P(+|\neg D) = 0.01$.

$$P(+) = 0.99 \times 10^{-4} + 0.01 \times 0.9999 \\ = 0.000099 + 0.009999 = 0.010098$$

$$P(D|+) = \frac{0.000099}{0.010098}$$

$$P(D|+) = 0.0098 \approx 0.98\%$$

- **Fewer than 1 in 100** people with a positive result actually have the disease.

“Why is it good news that the disease is rare”

- Because the **false positives outnumber the true ones**, and by a lot. In 1,000,000 people: **100** have the disease and 99 of them test positive; 999,900 do not, and **9,999** of them still test positive. **9,999 against 99 is roughly 100 to 1.**
- So a rare disease means a positive result is far more likely to be an error than a diagnosis. Rarity protects you.
- **Write the population count.** It is worth more marks than the algebra and it is much harder to get wrong.
- **The general lesson:** the posterior depends on the prior as much as on the test. A 99 % test is not a 99 % answer.

B6. The genetic defect 2+6

81 Ash 2+6 · part (b) asks for the **probability** that they do **not** have it, which is the complement – **read the question twice**

QUESTION AS PRINTED 2081 Ashwin · [2+6]

What are the approaches for measuring uncertainty? 1% of people have a certain genetic defect. 90% of tests for the gene detect the defect (true positives). 10% of the tests are false positives. a) What might be the probability of getting the genetic defect result? b) If a person gets a positive test result, what are the odds they actually don't have the genetic defect?

Solution

- **Given** $P(D) = 0.01$, $P(+|D) = 0.90$, $P(+|\neg D) = 0.10$.
- (a) The probability of a positive result:

$$\begin{aligned} P(+) &= 0.90 \times 0.01 + 0.10 \times 0.99 \\ &= 0.009 + 0.099 = \boxed{0.108 = 10.8\%} \end{aligned}$$

- (b)

$$\begin{aligned} P(\neg D|+) &= \frac{0.099}{0.108} \\ &= \boxed{0.9167 \approx 91.7\%} \end{aligned}$$

- And so $P(D|+) = 0.009/0.108 = 0.0833 = 8.3\%$.
 $91.7 + 8.3 = 100$. ✓

B7. Which factory made the defective bulb? 3+5

80 Ash 3+5 · **three-way partition**, so the denominator has three terms

QUESTION AS PRINTED 2080 Ashwin · [3+5]

How is reasoning done under uncertainty? Three factories produce light bulbs to supply the market. Factory A produces 20%, 50% of the bulbs are produced in factory B and 30% in factory C. 2% of the bulbs produced in factory A, 5% of the bulbs produced in factory B and 3% of the bulbs produced in factory C are defective. A bulb is selected at random in the market and found to be defective. What is the probability that this bulb was produced by factory B?

Solution

- **Given** $P(A) = 0.2$, $P(B) = 0.5$, $P(C) = 0.3$;
 $P(D|A) = 0.02$, $P(D|B) = 0.05$, $P(D|C) = 0.03$.
- **Total probability of a defective bulb**, one term per factory:

$$\begin{aligned} P(D) &= 0.2(0.02) + 0.5(0.05) + 0.3(0.03) \\ &= 0.004 + 0.025 + 0.009 = 0.038 \end{aligned}$$

$$P(B|D) = \frac{P(D|B)P(B)}{P(D)} = \frac{0.025}{0.038}$$

$$\boxed{P(B|D) = 0.6579 \approx 65.8\%}$$

B8–B9. The tall student, and the mammogram 3+5 2+6

78 Ch 3+5, 82 Ba 2+6

QUESTION AS PRINTED 2078 Chaitra · [3+5]

Explain how statistical reasoning aids in inference and reasoning in light of Bayes' theorem. At a certain University, 5% of men are over 6 feet tall and 2% of women are 6 feet tall. 60% of students are female. If a student is selected at random from among all those over six feet tall, what is the probability that the selected student is a woman?

QUESTION AS PRINTED 2082 Baishakh · CT 710 · [2+6]

What are casual networks? Given the following statistics, what is the probability that a woman has cancer if she has a positive mammogram result? i) One percent of women over 50 have breast cancer. ii) Ninety percent of women who have breast cancer test positive on mammograms. iii) Eight percent of women will have false positives.

Notes

- **As true odds**, which is what part (b) literally asks for: $0.099 : 0.009$, i.e. **11 to 1 against actually having the defect**. Give both the probability and the odds and neither reading can cost a mark.
- **Part (a) is ambiguous** as printed – “the probability of getting the genetic defect result” could mean $P(D)$, which is stated in the question and would make the part worthless. It means $P(+)$. Say which reading you took.
- Same shape as B5 with a much weaker test (10 % false positives instead of 1 %) and a much commoner condition (1 % instead of 0.01 %). The posterior lands at 8 % rather than 1 %.

Notes

- **The partition must be exhaustive:** $0.2 + 0.5 + 0.3 = 1$. ✓ Check it before you start; a partition that does not sum to 1 means a misread question.
- Factory B's share rises from **50 % to 66 %** because it is both the biggest producer *and* the worst. The other two posteriors are $P(A|D) = 0.004/0.038 = 10.5\%$ and $P(C|D) = 0.009/0.038 = 23.7\%$, and the three sum to 100 %. ✓
- **This is the template for every “which source did it come from” question**, and there are three of them in the archive (bulbs, factories, and the tall-student one below).

B8 the tall student

- Given $P(W) = 0.6$, $P(M) = 0.4$, $P(T|W) = 0.02$, $P(T|M) = 0.05$.

$$P(T) = 0.02(0.6) + 0.05(0.4) = 0.012 + 0.020 = 0.032$$

$$P(W|T) = \frac{0.012}{0.032} = \boxed{0.375 = 37.5\%}$$

- Women are the **majority** of students and the **minority** of tall ones: 60 % prior, 37.5 % posterior. **The evidence reverses the balance**, which is the sentence the 3-mark theory part wants.

B9 the mammogram

- Given $P(C) = 0.01$, $P(+|C) = 0.90$, $P(+|\neg C) = 0.08$.

$$P(+) = 0.90(0.01) + 0.08(0.99) \\ = 0.009 + 0.0792 = 0.0882$$

$$P(C|+) = \frac{0.009}{0.0882} = \boxed{0.1020 \approx 10.2\%}$$

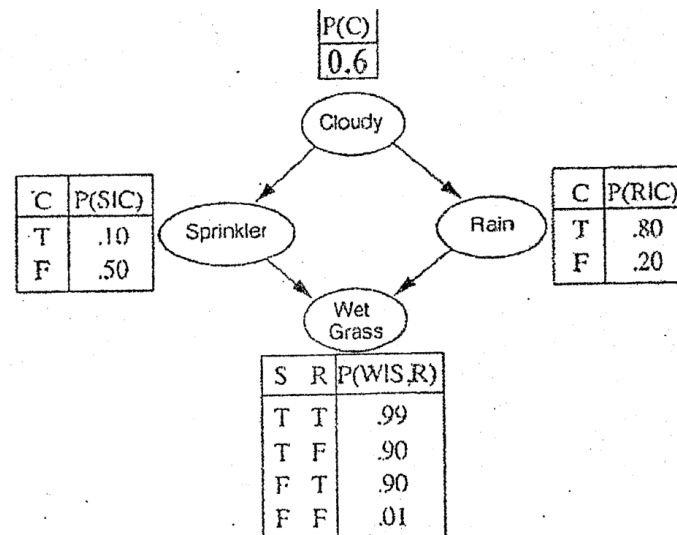
- **Nine positives in ten are false alarms**, even though the test catches 90 % of real cancers. Same base-rate lesson as B5, in the setting it was originally made famous in.
- **“Eight percent of women will have false positives”** means $P(+|\neg C) = 0.08$, not $P(\neg C \wedge +) = 0.08$. The looser reading gives $0.009/0.089 = 10.1\%$, close enough to hide the error – **state your reading**.

P8. Bayesian networks: reading a joint probability off the graph**B10. The wet grass network** 2+6

PIN 1/41 78 Ba 2+6 · the paper prints the network and its four CPTs · **one multiplication, four factors**

QUESTION AS PRINTED 2078 Baishakh · [2+6]

What do you understand by Bayesian Network? For the given Bayesian network below, find the probability of grass being wet when the weather is cloudy, there is some sprinkler but no rain, i.e. $C = \text{True}$, $S = \text{True}$, $R = \text{False}$ and $W = \text{True}$.



Solution

- **The product form**, one factor per node, each conditioned on its parents:

$$P(C, S, \neg R, W) = P(C) P(S|C) P(\neg R|C) P(W|S, \neg R)$$

- **Read each factor off its own table:**

- $P(C) = 0.6$ root node, prior
- $P(S|C=T) = 0.10$ sprinkler table, row T
- $P(R|C=T) = 0.80$, so $P(\neg R|C=T) = 1 - 0.80 = 0.20$
- $P(W|S=T, R=F) = 0.90$ wet-grass table, row T F

$$= 0.6 \times 0.10 \times 0.20 \times 0.90$$

$$P = 0.0108 = 1.08\%$$

The three ways this goes wrong

- **Forgetting to complement R .** The query says *no rain*, and the table gives $P(R|C)$. Take $1 - 0.80$. Using 0.80 gives 0.0432, four times too big, and it is the commonest wrong answer.
- **Picking the wrong CPT row.** The wet-grass table is indexed by (S, R) , and the query is $S = T, R = F$: row **T F**, value 0.90. Row **F T** is also 0.90, so a misread here happens to give the right number in this one query and the wrong one in every other.
- **Conditioning on the wrong parent.** Sprinkler and Rain both depend on *Cloudy* only; *WetGrass* depends on both of them and *not* on *Cloudy*. That is the conditional independence the network encodes, and it is what makes four numbers enough.
- **The 2-mark part:** a Bayesian network is a DAG of random variables with a CPT at each node, and the joint distribution is the product above. Six variables of this kind would need 31 free numbers as a full joint table; this one needs 9.

B11. Smoker → cancer → test 7

PIN 1/41 82 Bh 7 · asks for an *explanation*, but the marks are safest if the explanation carries numbers · a **chain** network, so every inference is a two-step

QUESTION AS PRINTED 2082 Bhadra · CT 710 · [7]

A Bayesian Belief Network (BBN) is used in a medical diagnosis system. The network includes the binary variables C (Patient has Cancer), T (Test result is Positive) and S (Patient is a Smoker), with structure $S \rightarrow C \rightarrow T$ and conditional probabilities $P(S=\text{Yes})=0.3$, $P(C=\text{Yes} | S=\text{Yes})=0.2$, $P(C=\text{Yes} | S=\text{No})=0.05$, $P(T=\text{Yes} | C=\text{Yes})=0.9$, $P(T=\text{Yes} | C=\text{No})=0.1$. Explain how the Bayesian Belief Network enables reasoning under uncertainty in this medical context.

The three kinds of reasoning, each with its number

- **Causal, top-down** – *a smoker walks in; how likely is a positive test?*

$$P(T|S) = 0.9(0.2) + 0.1(0.8) = \mathbf{0.26}$$

$$\text{against } P(T|\neg S) = 0.9(0.05) + 0.1(0.95) = 0.14.$$

- **Prior belief**, before anyone is seen:

$$P(C) = 0.2(0.3) + 0.05(0.7) = 0.06 + 0.035 = \mathbf{0.095}$$

$$P(T) = 0.9(0.095) + 0.1(0.905) = 0.0855 + 0.0905 = \mathbf{0.176}$$

- **Diagnostic, bottom-up** – *the test came back positive; how likely is cancer?* This is Bayes on the second edge:

$$P(C|T) = \frac{0.9 \times 0.095}{0.176} = \frac{0.0855}{0.176} = \mathbf{0.4858}$$

- **Through the chain** – *the test is positive; how likely is the patient a smoker?*

$$P(S|T) = \frac{P(T|S)P(S)}{P(T)} = \frac{0.26 \times 0.3}{0.176} = \frac{0.078}{0.176} = \mathbf{0.4432}$$

Consistency check: $P(T)$ computed through C gives 0.176, and computed through S gives $0.26(0.3) + 0.14(0.7) = 0.078 + 0.098 = 0.176$. ✓ **Two routes, one answer** – that is the network being coherent, and it is worth showing.

What to say in words, which is what the question asks

- The network **separates what is known from what is asked**. The doctor can only measure causal probabilities – how often smokers get cancer, how often cancer shows on a test – and the network stores exactly those. Bayes then runs them backwards on demand.
- It handles **partial evidence**: with no observation the belief in cancer is 9.5 %; knowing the patient smokes raises it to 20 %; a positive test raises it to **48.6 %**. Each piece of evidence revises, none replaces.
- **Conditional independence keeps it small**. T depends on C alone: once the cancer status is known, smoking tells you nothing more about the test. So three variables need **5 numbers** instead of $2^3 - 1 = 7$, and the saving grows exponentially with the network.
- **It never claims certainty**. A positive test leaves the patient more likely *not* to have cancer than to have it – 51.4 % against 48.6 % – which is precisely the kind of conclusion a logic-based KB cannot express at all.

B12. The three papers that ask for an example of your own 7 5+3 3+5

73 Bh 7, 80 Ba 5+3, 80 Ch 3+5 · no data printed, so **bring your own** — and the smallest one that shows everything is the two-node version of B9

The example to memorise

- **Network:** $Cancer \rightarrow Test$, two nodes, one edge.
- **CPTs:** $P(C) = 0.01$; $P(+|C) = 0.9$, $P(+|\neg C) = 0.08$.
- **Causal use** (down the arrow): a patient with cancer tests positive 90 % of the time. Straight from the table, no calculation.
- **Diagnostic use** (against the arrow): a positive test means cancer with probability

$$\frac{0.9 \times 0.01}{0.9(0.01) + 0.08(0.99)} = \frac{0.009}{0.0882} = 10.2\%$$

- **Three numbers, four lines, and it demonstrates the entire connection** the three papers ask about.

The connection, stated

- **The network supplies the numbers Bayes needs.** $P(C)$ is the prior sitting at the root; $P(+|C)$ and $P(+|\neg C)$ are the CPT of the child; the denominator is the law of total probability over that same CPT.
- **Bayes supplies the direction the network lacks.** The arrows point from cause to effect, which is the direction knowledge is available in. Diagnosis needs the other direction, and Bayes is the only thing that turns it round.
- So the answer to “*is there any connection*” is: **a belief network is a factored store of exactly the conditional probabilities Bayes’ theorem consumes**, and inference in the network is repeated application of Bayes plus the product rule.
- 80 Ba’s “how can inference be done” is answered by the four modes in §4.7: causal, diagnostic, intercausal and mixed. Name all four; work one.

How every answer above was checked**The proofs**

- `verify.py` carries a small **first-order resolution prover** (unification with occurs check, factoring, set of support) and runs it on **every clause set printed above**, in the exact form printed.
- It confirms the empty clause is derivable for all of them — and, just as importantly, that it is **not** derivable for the three papers whose printed goal does not follow:
 - 72 Ma *Shailendri likes chicken* — not derivable; *eats* is.
 - 82 Bh *Mr. A is a criminal* — not derivable; *Criminal(Mr.R)* is.
 - 79 Ash *Steve likes BK301* — not derivable from *likes* → *easy*; derivable from the converse.
- It also confirms that 69 Bh and 75 Ba **cannot** be closed without the two bridging axioms this document adds, and that they close once those are supplied.

The numbers

- Every Bayes posterior, every total-probability denominator and both network products are recomputed from the paper’s own figures and asserted against the boxed value above. **Nothing here was copied from a source.**
- Three of those checks are consistency tests rather than answers: the measles posteriors summing to 1, the three factory posteriors summing to 1, and 82 Bh’s $P(T)$ computed by two different routes through the network agreeing at 0.176.
- **Run it before trusting a rebuild:**
`cd AI/ExamNotes/src && python verify.py`
- A textbook error that survives a check is still an error. The three *Insights* corrections in this chapter — the split of a negated conjunction, the free variable where a Skolem function belongs, and the five-step CNF list — were each found by hand and then confirmed here.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
AI	artificial intelligence	ANI	artificial narrow intelligence
AGI	artificial general intelligence	ASI	artificial super intelligence
KR	knowledge representation	KRR	knowledge representation and reasoning
PEAS	performance measure, environment, actuators, sensors	GPS	General Problem Solver
NLP	natural language processing	NLU	natural language understanding
NLG	natural language generation	CAPTCHA	completely automated public Turing test to tell computers and humans apart
ML	machine learning	ANN, NN	artificial neural network
MLP	multilayer perceptron	BP	back-propagation
SOM	self-organizing map	GA	genetic algorithm
FIS	fuzzy inference system	ID3	Iterative Dichotomiser 3
CSP	constraint satisfaction problem	DFS	depth first search
BFS	breadth first search	DLS	depth limited search
IDS	iterative deepening search	UCS	uniform cost search
CNF	conjunctive normal form	WFF	well-formed formula
FOL	first-order logic	FOPL	first-order predicate logic
RRS	resolution refutation system	BBN	Bayesian belief network

Short	Means	Short	Means
ES	expert system	KB	knowledge base
CD	conceptual dependency	LISP	list processing (the language)
MOP	memory organization packet	isa	the class-to-class link of a semantic net (<i>instance</i> is the individual-to-class one)
attr(s)	attribute(s)	repo	repository
freq	frequency	config	configuration
info	information	no.	number
Eg	example	Vs	compare / differentiate
w/, w/o	with, without	+Fig	draw the figure
→	leads to, therefore	=	is, equals

Reading the year tags and chips

Exam-paper month codes: Ba = Baishakh · Jth = Jestha · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Po = Poush · Ma = Magh · Ch = Chaitra. Years are Bikram Sambat, so 80 Ch is 2080 Chaitra.

Bold year tags are Regular exams, plain are Back exams. A header printing “Regular / Back” in one box counts as Regular.

Three course codes, three faces. Plain = CT 653 (BCT, III-II, 26 papers) · monospace = CT 710 (BEI, IV-I, 7 papers) · *italic* = CT 78506 (BEX, IV-II, 8 papers). **The same year code names different papers under different codes:** 77 Ch and 77 Ch are two different exams, and only the font tells them apart.

Tier chips over all 41 papers, 2069–2082 BS: **TOP /41** = topic asked in **8 or more** papers · **HOT /41** = **4 to 7** papers · **PIN /41** = **1 to 3** papers but worth ≥ 6 marks.

Mark chips such as **2+6** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic never yet examined but written up anyway. **verified/added** marks anything researched and added that no source note in this folder covers.